

# Spatial prediction of soil organic carbon stocks across contrasting Andean basins, Peru

Carlos Carbajal<sup>a,\*</sup>, Merely Tumbalobos-Dextre<sup>a</sup>, Tatiana Condori-Ataupillco<sup>b</sup>,  
Nestor Cuellar-Condori<sup>c</sup>, Carla Gavilan<sup>d</sup>

<sup>a</sup> Dirección de Servicios Estratégicos Agrarios, Instituto Nacional de Innovación Agraria (INIA), Av. La Molina 1981, Lima 15024, Peru

<sup>b</sup> Dirección de Servicios Estratégicos Agrarios, Instituto Nacional de Innovación Agraria (INIA), Ayacucho 05002, Peru

<sup>c</sup> Dirección de Servicios Estratégicos Agrarios, Instituto Nacional de Innovación Agraria (INIA), Puno 21001, Peru

<sup>d</sup> Department of Environmental Sciences, Rutgers, The State University of New Jersey, New Brunswick, NJ 08904, USA

## ARTICLE INFO

### Keywords:

Digital soil mapping  
Soil organic carbon stock  
Geographically weighted regression  
Machine learning regression algorithms  
Andes

## ABSTRACT

Soil organic carbon stocks (SOCS) are critical components of the global carbon cycling and play a central role in climate change mitigation. However, their dynamics in high-altitude Andean ecosystems remain poorly understood despite their importance for carbon sequestration. The significant spatial heterogeneity of SOCS in mountainous terrain makes accurate quantification and mapping challenging. This study evaluated the performance of geospatial regression and machine learning (ML) approaches for predicting SOCS in two Peruvian Andean basins: Torobamba and Coata. We compared Geographically Weighted Regression (GWR), GWR with collinearity analysis (GWRC), their kriging-adjusted variants, and ML models (Random Forest, Gradient Boosting). Models were built using key SOCS covariates for each basin and validated through 5-fold cross-validation with Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and coefficient of determination ( $R^2$ ). In Torobamba, GWRC markedly improved performance, reducing the RMSE by 79–90 % and achieving  $R^2$  up to 0.99. In contrast, Coata, showed only modest improvements (RMSE reductions of 7.8–9.8 %,  $R^2 = 0.30$ –0.39). ML models performed poorly (negative  $R^2$ ), likely due to feature selection, parameter tuning, or limited sample size. Overall, locally weighted regression approaches (GWRK/GWRCK) outperformed conventional ML methods for SOCS prediction in complex mountain environments, particularly with small to medium sample sizes. These results highlight the importance of accounting for spatial non-stationarity in SOCS and provide methodological guidance for SOCS mapping in Andean ecosystems.

## 1. Introduction

Digital soil mapping (DSM) has become an essential tool for understanding and predicting soil properties in diverse landscapes (Chen et al., 2022). In the Andean region, a complex terrain with significant ecological diversity, soil organic carbon (SOC) is critical for soil health, productivity, and ecosystem sustainability (Munoz et al., 2015). Traditional soil sampling methods, although effective at local scales, are often labor-intensive, time-consuming (Minasny and McBratney Alex, 2016), and limited in their ability to provide comprehensive information of SOC distribution across vast, heterogeneous areas (Minasny et al., 2013). SOC is vital for maintaining soil structure, enhancing water retention, and supporting biodiversity, functions particularly important in steep, erosion-prone Andean landscapes. However, SOC levels are

spatially variable and challenging to measure across mountainous topography (Román-Sánchez et al., 2018). DSM enables the generation of continuous SOC maps that capture the variability of soil properties driven by climate, vegetation, land use, and topography (Moura-Bueno et al., 2021). This approach is well suited for the Andes, where mountainous landscapes and diverse microclimates drive to distinct soil formation processes (Anderson et al., 2011).

To monitor and quantify C sequestration, SOC concentrations are expressed in terms of volume and are referred to as soil organic carbon stocks (SOCS). Several studies investigated the spatial distribution of SOCS under different land use and land cover (LULC) types in the montane ecosystem (Dorji et al., 2014) and forests (Ottoy et al., 2017; Vallejos-Torres et al., 2024). Other have assessed SOCS at various depths using environmental covariates, and compared them across soil types

\* Corresponding author.

E-mail addresses: [cmcarbajal@gmail.com](mailto:cmcarbajal@gmail.com) (C. Carbajal), [cg1141@envsci.rutgers.edu](mailto:cg1141@envsci.rutgers.edu) (C. Gavilan).

<https://doi.org/10.1016/j.geodrs.2025.e01026>

Received 29 April 2025; Received in revised form 30 October 2025; Accepted 5 November 2025

Available online 6 November 2025

2352-0094/© 2025 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

and land uses, reporting that SOCS increase under grasslands (Hengli et al., 2023). Understanding SOCS distribution in the high Andean regions of Peru is constrained by restricted access and logistical challenges due to the complex geography, further limiting the sample size; moreover, if observations at different depths are required.

Predicting the current or future spatial distribution of SOCS is possible using statistical and ML models (Yigini and Panagos, 2016). In the present study, we focused on the performance of Geographically Weighted Regression (GWR), which employs local models estimated from subsets of observations centered at a focal point (Szakács et al., 2011). GWR has consistently outperformed other models, such as ordinary least squares (OLS), when estimating the spatial distribution of SOCS in forests, effectively reducing data saturation and uncertainty (Wu et al., 2023). Compared to Ordinary Kriging (OK), inverse distance weighted (IDW), and multiple linear regression (MLR), GWR produced plausible results, revealing spatial patterns affected by environmental variables (Zhang et al., 2011). GWR is usually applied to address multicollinearity effects among auxiliary variables in regression prediction and to capture effectively spatially non-stationary structures and relationships between the target and response variables (Chen et al., 2021; Guo et al., 2018). Future studies should examine how GWR models address multicollinearity while systematically evaluating their predictive performance under varying bandwidth selection, kernel functions, and spatial autocorrelation patterns. For example, GWR Kriging (GWRK) proved more accurate than standard GWR when used to correct for residual autocorrelation (Kumar et al., 2012). Similarly, the potential benefits of integrating machine learning require further investigation (Zhang et al., 2025). For instance, Zeng et al. (2024) developed hybrid models integrating GWR and ML approaches, such as geographically weighted neural networks, to predict Cu distributions in urban topsoil. In other instances, the method was combined with RF to develop spatially weighted RF models, which were used to investigate the drivers of forest dynamics (Santos et al., 2019). For similar purposes, the strengths of Bayesian inference were leveraged through the Bayesian Geographically Weighted Regression model (Faisal et al., 2025).

Advancements in ML, particularly random forest (RF) and gradient boosting (GB) techniques, offer alternatives for the efficient and precise prediction of SOCS. RF is an ensemble learning method that constructs multiple decision trees during training and combines them to improve predictive accuracy and control over-fitting (Breiman, 2001). It is particularly effective for handling large datasets with complex interactions among variables (Genuer et al., 2017; Schonlau and Zou, 2020). RF leverages environmental, topographic, and climatic information from remote sensors and spectral indices to generate detailed SOC maps (Pouladi et al., 2023; Triantakoustantis and Karakostas, 2025), with good results in grasslands (Szakács et al., 2011). A notable application of RF is the creation of a gridded SOCS map across Australia, which enhanced the representation of uncertainties distribution (Wang et al., 2024). GB, another ensemble learning method, builds multiple decision trees sequentially, with each tree correcting the errors of the previous one (Friedman, 2001; Zhang and Jung, 2021). This approach results in a highly accurate predictive model particularly effective in capturing complex patterns and relationships within data (Habib et al., 2024; Tahmouresi et al., 2024), making GB suitable for SOCS prediction. In previous research, GB models were used environmental covariates, topographic information, and remote sensing data to estimate SOC with high precision (Chen et al., 2024; Kumar et al., 2023; Taghizadeh-Mehrjardi et al., 2020). ML model accuracy depends on input data quality and availability. While incomplete or erroneous data can compromise performance, the iterative nature of GB allows continuous model refinement. This characteristic makes GB a reliable approach for predicting SOC across diverse landscapes (Taghizadeh-Mehrjardi et al., 2020). Nevertheless, ensemble methods such as RF and GB face two main limitations: high computational demands and poor interpretability due to their complex models. These factors can hinder its adoption in practical applications (Halder et al., 2024). The effectiveness of ML

depends on data quality, predictor selection and parameter tuning. Combining deep sampling with hybrid models can enhance interpretability and predictive power, while prioritizing the selection of appropriate predictors for the specific study area is also recommended.

This study aims to a) develop and evaluate advanced predictive frameworks, specifically focused on GWR models and ML algorithms, to predict SOCS across two contrasting basins in the Andean region of Southern Peru, b) investigate the influence of environmental and spatial patterns on the distribution of SOCS in mountain ecosystems, and c) provide guidance on land management to improve SOC storage.

## 2. Materials and methods

### 2.1. Study area

Two study areas were selected for the present study: the Torobamba and Coata basins (hereafter referred to as Torobamba and Coata), located in the southern Peruvian region (Fig. 1). Torobamba ( $-74^{\circ}0.16' \text{ N}$ ,  $-13^{\circ}0.20' \text{ E}$ ), with an area of approximately  $1040.47 \text{ km}^2$ , is located in the Peruvian Andean southern region (between 1500 and 4500 masl) and comprises extensive agricultural areas, shrublands, native grasslands, and seasonal peatlands and bogs (Zanaga et al., 2022). The mean temperature ranges between  $1^{\circ} \text{ C}$  to  $20^{\circ} \text{ C}$  during the dry season (May to September) and between  $4^{\circ} \text{ C}$  to  $20^{\circ} \text{ C}$  during the wet season (October to April), with annual precipitation of 536 mm. Soils in this basin are classified mainly as Cambisols, and other soils reported in the area are Leptosols, Andosols, Regosols, and Vertisols (IUSS Working Group, 2007). Coata ( $-70^{\circ}0.29' \text{ N}$ ,  $-15^{\circ}0.62' \text{ E}$ ), with an area of approximately  $460.16 \text{ km}^2$ , is located in the southern Peruvian region between 3825 and 4250 masl. This basin comprises extensive agricultural areas and native grasslands presenting soils classified as Cambisols, Chernozems, and Phaeozems (IUSS Working Group, 2007). The historical average annual precipitation is 751 mm, average minimum temperatures of  $-5^{\circ} \text{ C}$ , and average maximum temperatures of  $17^{\circ} \text{ C}$ . Both basins exhibit distinct biogeographic characteristics based on an ecological classification system proposed by Pulgar Vidal (2014). Torobamba demonstrates high environmental heterogeneity, spanning four ecological zones: Yunga (warm-dry), Quechua (temperate, dry and pleasant), Suni (cold-dry), and Puna (very cold). However, Coata is only found in the Puna zone, which is known for being cold, semi-humid to humid environments and natural grassland. The Puna ecosystem plays a crucial role in soil carbon sequestration throughout the Southern Cone, particularly due to the presence of peatlands and organic soil systems (García Lino et al., 2024).

### 2.2. Soil sampling and analysis

The conditional Latin Hypercube Sampling (cLHS) design (Minasny and McBratney, 2006) was used to find the best distribution of sampling points across the study area. The cLHS was implemented using the 'cLhs' R package (Roudier, 2012). A set of environmental variables was used as conditioning factors to generate sampling points using the cLHS method (Table S1). A total of 100 and 96 sampling sites were identified for Torobamba and Coata, respectively. The geographic coordinates were extracted for each of the sampling sites.

At each identified sampling site, five soil subsamples were collected within a radius of 5 m at a depth of 30 cm. The subsamples were homogenized to obtain approximately 1 kg of composite sample per site. Undisturbed samples were collected at 30 cm depth using a 5 cm by 5 cm core ring sampler in order to assess soil bulk density (BD) at each identified sampling site.

The SOC (%) was determined by dry combustion (ISO, 1996) using an elementary analyzer LECO CN828 (Leco Corp., St. Joseph, MI, USA), and BD was obtained using the core method (ISO, 2017). Eq. (1) (Peng et al., 2024), SOCS at 30 cm were determined by calculating soil mass ( $\text{BD} \times \text{soil depth}$ ) per unit area and multiplying by SOC concentration

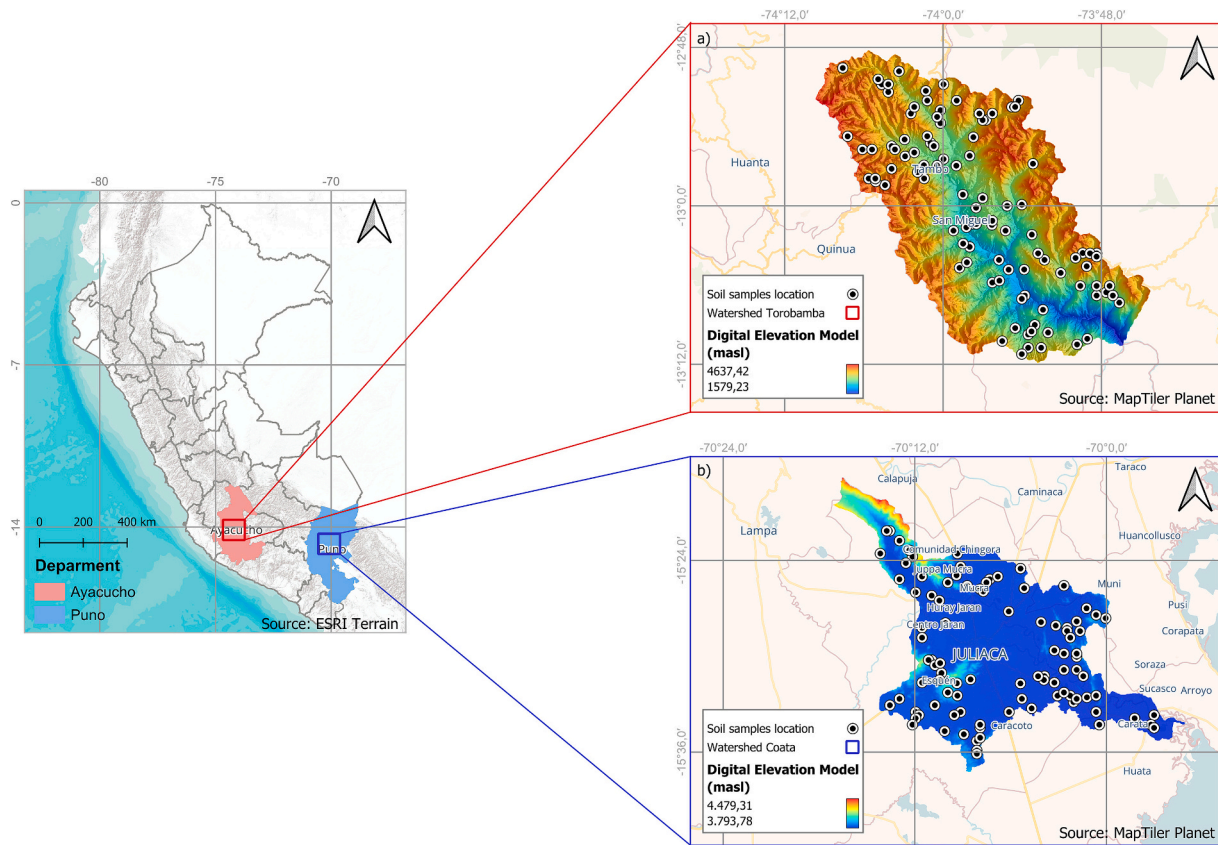


Fig. 1. Location of sampling points in a) Ayacucho and Puno Regions of Peru with the respective basins, b) Torobamba, and c) Coata.

(%).

$$\text{SOCs (Mg ha}^{-1}\text{)} = \text{SOC (\%)} \times \text{BD (Mg m}^{-3}\text{)} \times \text{Soil Depth (cm)} \quad (1)$$

### 2.3. Covariates data collection

Multiple environmental factors influence SOC distribution at a

landscape level. To model the response of SOCS, it is essential to identify relevant predictor covariates. The foundation of this approach was laid out by Jenny (1994) and the SCORPAN paradigm of soil forming factors (McBratney et al., 2003). A total of 21 covariates were considered in this study (Table S2). A set of topographic variables was derived from the ALOS PALSAR digital elevation model (DEM) at 30 m resolution using SAGA tools (Conrad et al., 2015) and the ‘elevatr’ R package (Hollister

Table 1

List of the indices derived from remote sensing used for spatial modeling.

| Index                                        | Acronym | Equation                                                           | Details                                                                                                                                                                                                                    | Reference               |
|----------------------------------------------|---------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Normalized Difference Vegetation Index       | NDVI    | $\frac{(NIR - R)}{(NIR + R)}$                                      |                                                                                                                                                                                                                            | (Rouse et al., 1974)    |
| Transformed Soil Adjusted Vegetation Index   | TSAVI   | $\frac{s(NIR - s^*R - a)}{(a^*NIR + R - a^*s + X^*(1 + s^2))}$     | $s$ = Soil line slope<br>$a$ = Soil line intercept<br>$X$ = Adjustment factor to minimize soil noise                                                                                                                       | (Baret and Guyot, 1991) |
| Soil Adjusted Total Vegetation Index         | SATVI   | $\frac{SWIR1 - R}{SWIR1 + R + L} \times (1 + L) - \frac{SWIR2}{2}$ | $L$ = Soil brightness factor (values between 0 and 1)                                                                                                                                                                      | (Marsett et al., 2006)  |
| Green Normalized Difference Vegetation Index | GNDVI   | $\frac{(NIR - G)}{(NIR + G)}$                                      |                                                                                                                                                                                                                            | (Gitelson et al., 1996) |
| Enhanced Vegetation Index                    | EVI     | $2.5 \times \frac{(NIR - R)}{(NIR + C1^*R - C2^*B + L)}$           | $L$ = Canopy Background Adjustment (values between 0 and 1)<br>$C1$ = Aerosol Resistance Coefficient in band Red (values between 0 and 6)<br>$C2$ = Aerosol Resistance Coefficient in band Blue (values between 0 and 7.5) | (Huete et al., 2002)    |
| Soil Adjusted Vegetation Index               | SAVI    | $\frac{(NIR - R)}{(NIR + R + L)} \times (1 + L)$                   | $L$ = Soil brightness factor (values between 0 and 1)                                                                                                                                                                      | (Huete, 1988)           |
| Normalized Difference Water Index            | NDWI    | $\frac{(NIR - SWIR1)}{(NIR + SWIR1)}$                              |                                                                                                                                                                                                                            | (Gao, 1996)             |
| Bare Soil Index                              | BSI     | $\frac{(SWIR1 + R) - (NIR + B)}{(SWIR1 + R) + (NIR + B)}$          |                                                                                                                                                                                                                            | (Rikimaru et al., 2002) |
| Second Brightness Index                      | BI2     | $\frac{\sqrt{R^2 + G^2 + NIR^2}}{3}$                               |                                                                                                                                                                                                                            | (Escadafal, 1989)       |

et al., 2023). Climate variables, including annual mean temperature and precipitation, were extracted from the WorldClim v2.1 database at 1 km resolution (Fick and Hijmans, 2017) using the ‘geodata’ R package (Hijmans et al., 2021). Land Surface Temperature (LST) was incorporated as a critical climatic indicator governing soil-atmosphere interactions and moisture dynamics (Ghaderpour et al., 2024). Following the method described in (Parastatidis et al., 2017), Google Earth Engine (GEE) (Gorelick et al., 2017) was used to retrieve LST by using Landsat 8 imagery, including the thermal band (ST\_B10) and the NDVI (Normalized Difference Vegetation Index).

A suite of vegetation indices (Table 1) was produced from USGS Landsat 8 Level 2, Collection 2 imagery (30 m resolution) acquired through GEE. Several indices, such as NDVI, TSAVI (Transformed Soil Adjusted Vegetation Index), SATVI (Soil Adjusted Total Vegetation Index), GNDVI (Green Normalized Difference Vegetation Index), EVI (Enhanced Vegetation Index), SAVI (Soil Adjusted Vegetation Index), NDWI (Normalized Difference Water Index), BSI (Bare Soil Index), and BI2 (Second Brightness Indices), were obtained and included in the analysis.

## 2.4. Statistical analysis and data preprocessing

A structured feature selection process was implemented in R software 4.4.3 (R Core Team, 2023) to enhance model efficacy and mitigate overfitting. The process involved two sequential steps: first, address redundancy by removing variables exhibiting correlations exceeding 90 %, and second, iteratively eliminate features with Variance Inflation Factors (VIFs) above a threshold of 10. Prior research (Naimi et al., 2021; Zhang et al., 2022) suggests discarding variables with high VIF (>10) to enhance model robustness and prediction accuracy. This threshold is considered a thumb rule, and should be treated with caution when used to assess signals of persistent multicollinearity (O’Brien, 2007). The VIF was computed according to Eq. (2). The procedure retained predictors that met the VIF criteria, as illustrated in the first block of Fig. 2. Correlation matrix heatmaps are provided in the supplementary materials (Figs. S1 and S2).

$$VIF_i = \frac{1}{1 - R_i^2} \quad (2)$$

where  $R_i^2$  is the R-squared value obtained from the regression of  $X_i$  on the other predictors.

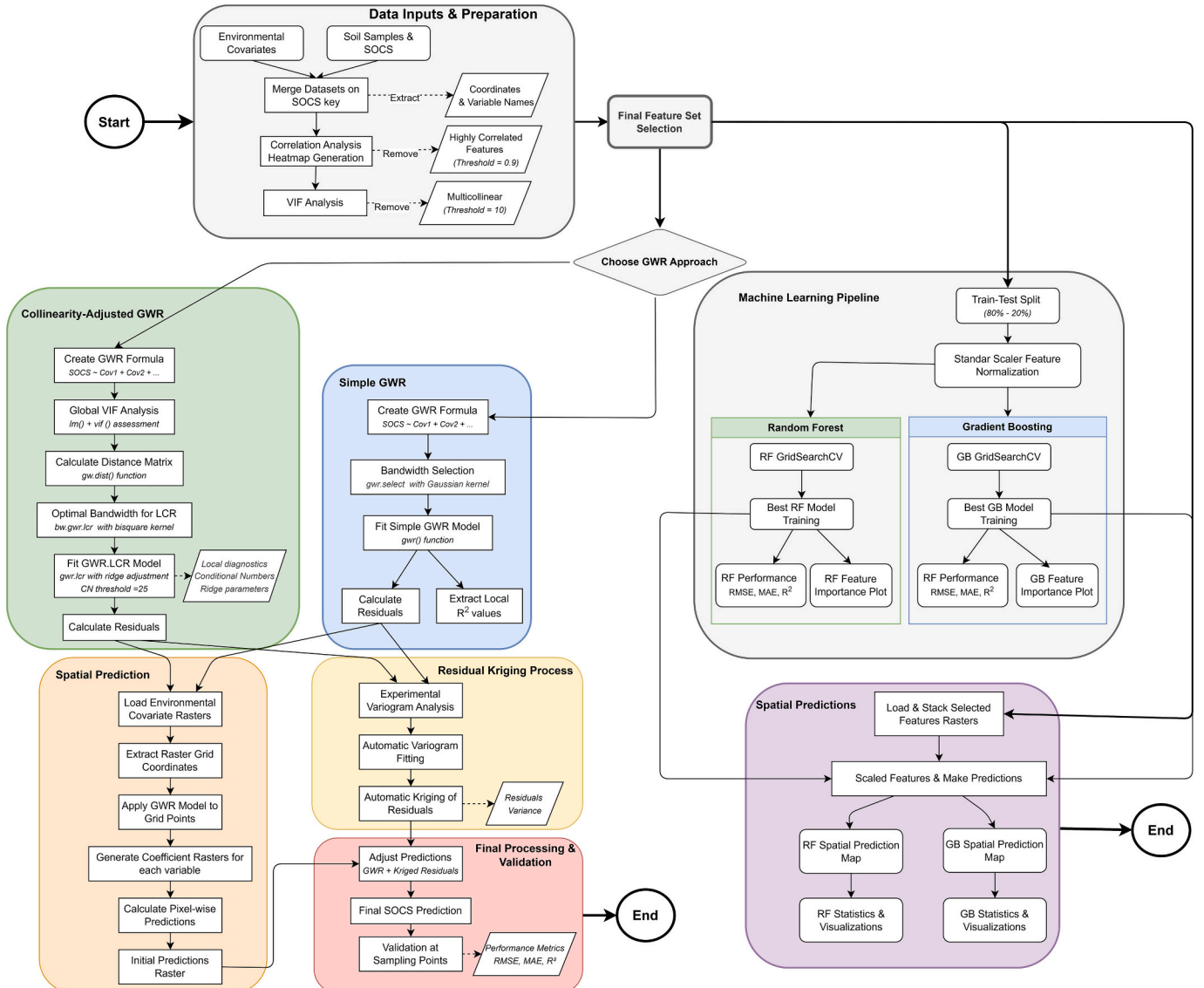


Fig. 2. Schematic representation of the modeling framework utilizing Geographically Weighted Regression and Machine Learning approaches.

## 2.5. Modeling SOCS

The present study uses Geographically Weighted Regression (GWR), Random Forest (RF), and Gradient Boosting (GB) models to predict SOCS. These models were trained using a dataset of environmental predictors selected during preprocessing, including surface soil properties, topographic and climatic variables, as well as vegetation spectral indices. The approaches assumed spatially variable relationships between SOCS and environmental covariates. The GWR model explicitly addressed this relationship in its localized regression framework (Zeng et al., 2016). To complement the GWR characteristics, ML models such as RF and GB were also implemented to predict SOCS by synthesizing and identifying diverse covariate groups, balancing complexity and generalizability (Hounkpatin et al., 2021; Ottoy et al., 2017).

### 2.5.1. Geographically weighted regression (GWR)

GWR is a local spatial analysis method that explores relationships between predictor variables, often addressing spatial heterogeneity or non-stationarity (Brunsdon et al., 1996). This technique generates localized regression results at any point in a region by using statistical curve-fitting and smoothing techniques (Thrift and Kitchin, 2009). Unlike global regression outputs, it provides mappable statistics that describe local relationships (Fotheringham et al., 1998). The following Eq. (3) presents an adaptation of the GWR model (Emamgholizadeh et al., 2017) that is used in this study to predict SOCS.

$$y(u) = \beta_0(u) + \sum_{k=1}^p \beta_k(u) X_k(u) + \epsilon(u) \quad (3)$$

where  $y(u)$  represents the dependent variable (SOCS) at location  $u$ ,  $\beta_0(u)$  is the geographically varying intercept,  $\beta_k(u)$  are the unknown regression coefficients that are spatially variant,  $X_k(u)$  are the  $k$ th independent variables with the value  $X_k$  at location  $u$ ,  $p$  is the total number of samples of soils, and  $\epsilon(u)$  represents the random error at location  $u$ .

GWR analysis began by extracting coordinates to determine spatial relationships. The model incorporated all variables except SOCS (dependent variable). The optimal bandwidth selection was obtained by employing cross-validation with a Gaussian weighting kernel using the 'gwr.sel' function from the 'spgwr' R package (Bivand and Yu, 2006). The GWR model, fitted with optimal bandwidth, included hat matrix calculations for diagnostics and generated location-specific  $R^2$  values to identify spatial variations in model performance. Prediction layers were built using the extracted coordinates from the selected covariates, including the coefficient and intercept values. Model refinement employed kriging interpolation of residuals using the 'autofitVariogram' and 'autoKrig' functions from the automap R package (Hiemstra et al., 2009), resulting in adjusted predictions.

### 2.5.2. Geographically weighted regression with collinearity analysis (GWRC)

To enhance model robustness and provide a reference point for comparing local collinearity patterns, global multicollinearity was assessed using Variance Inflation Factors (VIF), followed by localized collinearity analysis incorporating coordinate extraction and distance matrix computation. The distance matrix in GWR serves as a crucial element, as it records the pairwise distances between all data points based on their coordinates. The optimal spatial bandwidth for the GWR framework was determined via *bisquare* kernel and adaptive bandwidth selection to determine the number of nearest neighbors using the 'bw.gwr.lcr' function within the GWmodel package (Gollini et al., 2015). The GWRC model incorporated a locally compensated ridge term (*gwr.lcr*) to mitigate spatially varying multicollinearity. A Local Conditional Number (CN) threshold of 25 was used to identify locations requiring local ridge parameter adjustment. Tuning the Lambda parameter (*lambda.adjust*) further stabilized the regression coefficients at these sites. This approach reduces their variance and stabilizes the model (Wheeler, 2007). Diagnostic plots were generated to visualize CN, a

measure of the local multicollinearity among the independent variables at each location, and ridge parameters. Also, predictions were extrapolated across the study areas, ensuring spatially explicit outputs.

### 2.5.3. Random Forest (RF)

Random Forests (RF), which leverage the principles of classification and regression trees, bagging, and added randomness, are robust prediction tools that enhance accuracy through the law of large numbers while mitigating overfitting (Breiman, 2001; Tyralis et al., 2019). With selected variables during the preprocessing, the data were partitioned into training (80 %) and testing (20 %) subsets, standardized to ensure consistent scaling. A grid search was implemented using the function 'GridSearchCV', systematically testing hyperparameter combinations on scaled training data to optimize the performance of the 'RandomForestRegressor' model with 5-fold cross-validation. The process culminated in identifying the best-performing model, with the optimal hyperparameters that maximized predictive accuracy and generalization capability. This approach ensures methodological rigor in balancing model complexity and validation robustness.

### 2.5.4. Gradient boosting (GB)

Gradient Boosting (GB) is an ensemble ML method that builds predictive models by iteratively combining weak base learners (e.g., decision trees) to minimize residuals through a forward stage-wise optimization process (Belyadi and Haghighat, 2021). Designed for regression tasks, GB operates by sequentially fitting regression trees to the negative gradient of a differentiable loss function, incrementally improving predictions for continuous target variables (Pedregosa et al., 2011). This additive modeling approach balances bias-variance trade-offs, enabling robust performance in complex regression scenarios while maintaining computational efficiency.

After splitting and scaling out the data, the 'GradientBoostingRegressor' method applied to the train data, and the hyperparameter optimization of the model was performed using the function 'GridSearchCV', to evaluate various combinations of key parameters, such as the number of estimators (100, 200, 500), learning rate (0.01, 0.1, 0.2), maximum tree depth (3, 4, 5), minimum samples required to split a node (2, 5, 10), and minimum samples required in a leaf node (1, 2, 4). This optimization procedure used a 5-fold cross-validation and employed a negative mean squared error as the performance metric, leveraging parallel processing to improve computational efficiency.

## 2.6. SOCS and land cover

To analyze the spatial distribution of SOCS across different land cover (LC) types in the study area, a supervised classification was performed using Sentinel-2 imagery within the GEE platform (Gorelick et al., 2017). Training points were incorporated for representative sampling. Prediction results from each model were intersected by LC type; the mean statistics of training points were calculated to show the SOCS variation. LC types in the legend are based on the ESA WorldCover 10 m 2021 v200 thematic map (Zanaga et al., 2022). The LC maps for each basin are available in the supplementary material (Fig. S3).

## 2.7. Model validation

The accuracy assessment of the preliminary models involved calculating residuals from predicted and observed values. Kriging via residual interpolation was used to adjust only the GWR predictions. We then used metrics such as the root square error (RMSE), mean absolute error (MAE), and coefficient of determination ( $R^2$ ) for a thorough model performance assessment, as described in Eqs. (4)–(6), respectively.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (5)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (6)$$

where  $y_i$  are the observed values,  $\hat{y}_i$  are the predicted values,  $\bar{y}$  the mean of the observed values, and  $n$  is the number of samples.

### 3. Results

#### 3.1. Exploratory data analysis

A descriptive statistical summary of topsoil (0–30 cm) SOC and BD for both study areas is provided in Table 2. The average SOC values for Torobamba and Coata were 2.52 % ( $\pm 1.04$ ) and 1.34 % ( $\pm 0.39$ ), respectively. Torobamba exhibited significantly higher values of SOC and heterogeneity (0.83–5.16 %). Average BD values were 1.24 Mg m<sup>-3</sup> ( $\pm 0.19$ ) for Torobamba and 1.32 Mg m<sup>-3</sup> ( $\pm 0.17$ ) for Coata. Torobamba has the highest SOC values and the lowest BD values, unlike Coata, which has the lowest SOC values but the highest BD values.

Two tables in the supplementary materials provide descriptive statistics for each basin based on the study covariates and the response variable SOCS (Tables S3 and S4).

#### 3.2. Selected covariates

Table 3 shows the selected covariates for each basin based on the VIF criteria. Nine covariates were selected to model and predict SOCS. Multicollinearity among independent variables in the regression models, measured by the VIF, varied between 1.39 and 8.67 for Torobamba and 1.07 and 8.89 for Coata.

#### 3.3. GWR model analysis and predictions

The ample coefficient range displayed by the GWR models lead to significant spatial variation in the relationships between the selected covariates and the SOCS predictions (Table 4). The GWR model achieves an optimal bandwidth of 55,023.57 m; however, GWRC model achieves an optimal bandwidth of 63 neighbors for Torobamba. For Coata, the GWR model has a fixed optimal bandwidth of 40,064.74 m, while the GWRC has an optimal bandwidth of 95 neighbors.

Spatial variation in the local  $R^2$  values for the GWR models is minimal (around 1–2 %). Across locations in Torobamba, the  $R^2$  values fell within a range of 0.133 to 0.141. Coata shows similar results, with  $R^2$  values between 0.164 and 0.189. In both basins, the higher  $R^2$  values were located in the southeast region (Fig. 3). The model explains only 13–14 % and 16–19 % of SOCS variation, which means that the GWR model is consistent but has weak local prediction power across the study areas.

When applying Geographically Weighted Ridge Regression (GWRR) in GWRC, we obtained the spatial distribution of CN, showing values

**Table 2**

Descriptive statistics values of Soil Organic Carbon (SOC) and bulk density (BD) at the sampling sites in Torobamba and Coata.

| Basin     | Variable | Units              | Mean | SD <sup>1</sup> | Min <sup>2</sup> | Max <sup>3</sup> |
|-----------|----------|--------------------|------|-----------------|------------------|------------------|
| Torobamba | SOC      | %                  | 2.52 | 1.04            | 0.83             | 5.16             |
|           | BD       | Mg m <sup>-3</sup> | 1.24 | 0.19            | 0.65             | 1.59             |
| Coata     | SOC      | %                  | 1.34 | 0.39            | 0.52             | 2.42             |
|           | BD       | Mg m <sup>-3</sup> | 1.32 | 0.17            | 0.71             | 1.75             |

<sup>1</sup> Standard deviation.

<sup>2</sup> Minimum.

<sup>3</sup> Maximum.

**Table 3**

Variance Inflation Factor (VIF) values for selected covariates in the two study areas.

| Torobamba                              |      | Coata                                        |      |
|----------------------------------------|------|----------------------------------------------|------|
| Covariate                              | VIF  | Covariate                                    | VIF  |
| Soil Adjusted Total Vegetation (SATVI) | 8.67 | Valley Depth (V_depth)                       | 8.89 |
| Normalized Height (N_height)           | 7.78 | Soil Adjusted Total Vegetation Index (SATVI) | 8.89 |
| Second Brightness Index (BI2)          | 6.00 | Topographic Wetness Index (TWI)              | 8.54 |
| Valley Depth (V_depth)                 | 5.61 | Normalized Height (N_height)                 | 6.68 |
| Length Slope Factor (LS_factor)        | 5.26 | Bare Soil Index (BSI)                        | 6.17 |
| Topographic Position Index (TPI)       | 2.88 | Length Slope Factor (LS factor)              | 2.53 |
| Plan Curvature (PLcurv)                | 2.71 | Topographic Position Index (TPI)             | 2.21 |
| Profile Curvature (PRcurv)             | 2.45 | Profile Curvature (PRcurv)                   | 1.51 |
| Bare Soil Index (BSI)                  | 1.39 | Plan Curvature (PLcurv)                      | 1.07 |

between 17.77 and 27.33 for Torobamba and 20.11 and 26.70 for Coata (Fig. 4).

The predicted SOCS using GWR with adjusted residuals ranged from 37.31 to 173.97 (Mg ha<sup>-1</sup>) in Torobamba and from 10.99 to 100.21 (Mg ha<sup>-1</sup>) in Coata. The kriged-interpolated residuals in the GWRC model presented values ranging from 27.84 to 161.01 (Mg ha<sup>-1</sup>) in Torobamba and from 13.95 to 95.35 (Mg ha<sup>-1</sup>) in Coata (Fig. 5).

#### 3.4. Machine learning models analysis

Variable importance (VI) plots show the contribution of covariates to the accuracy of the prediction model (Fisher et al., 2019). For the RF models, variables such as V\_depth, SATVI, BI2, TPI, and BSI were identified as the most important in Torobamba, while V\_depth, BSI, SATVI, N\_height, and PRcurv were the most important in Coata (Fig. 6).

For the GB models, topographic variables such as TPI, LS\_factor, and V\_depth, and vegetation-related variables such as BSI and SATVI, were the most important in Torobamba. In Coata, V\_depth, TWI, N\_height, and PLcurve were the most important topographic variables, while only SATVI represents the vegetation-related covariates (Fig. 7).

The spatial distribution of SOCS predicted by RF and GB models is illustrated in Fig. 8. Torobamba exhibited high SOCS values, ranging from 71.8 to 122.95 Mg ha<sup>-1</sup> (RF) and 68.3 to 124.72 Mg ha<sup>-1</sup> (GB), compared to Coata, which showed values between 40.02 and 69.9 Mg ha<sup>-1</sup> (RF) and 37.29 to 75.02 Mg ha<sup>-1</sup> (GB). While both models yielded comparable predictions, GB demonstrated a more uniform spatial distribution pattern, whereas RF showed more localized values.

#### 3.5. Change in SOCS by land cover

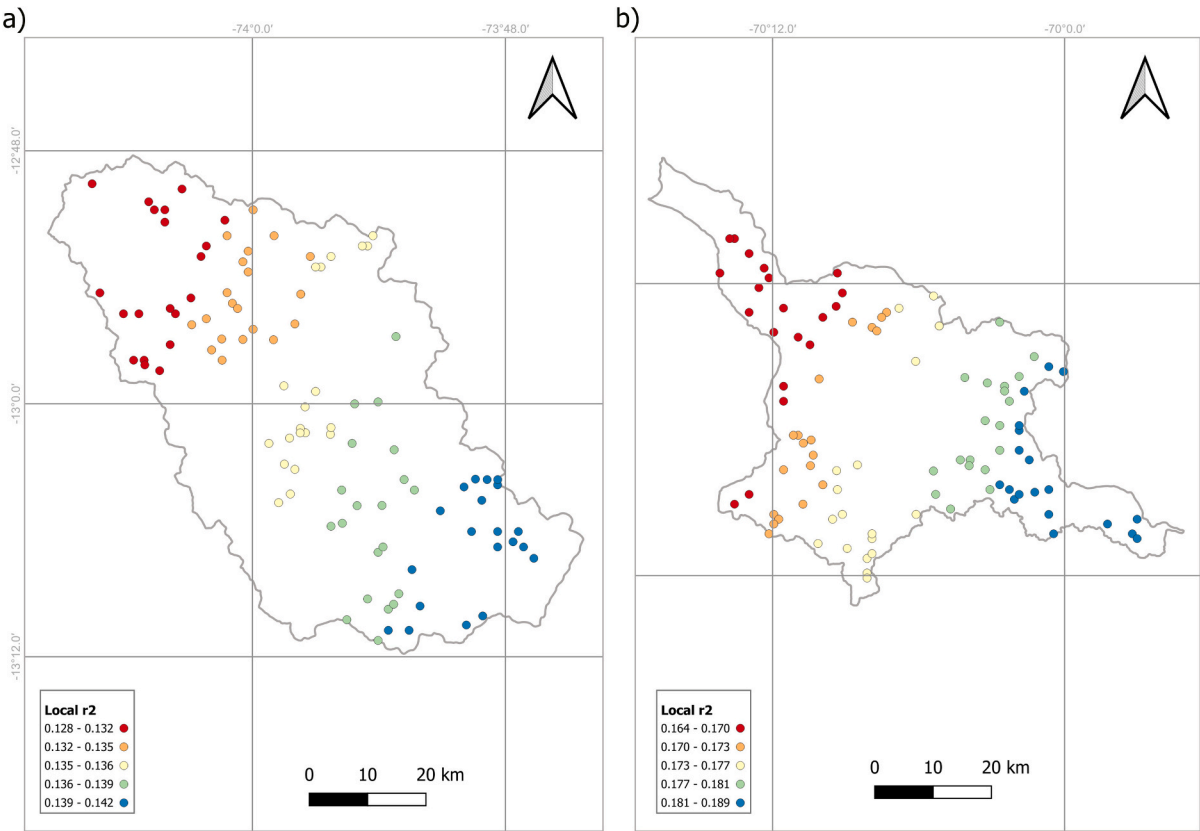
The predicted SOCS grouped by their corresponding LC are shown in Fig. 9 and Fig. 10. Torobamba exhibits the highest values of SOCS in the Shrubland cover (95.79 Mg ha<sup>-1</sup>) predicted by the GWRKC model, followed by the Tree cover (94.17 Mg ha<sup>-1</sup>) using the GWRK model. Cropland has the lowest values when using GWRKC (85.88 Mg ha<sup>-1</sup>) and GWRK (86.38 Mg ha<sup>-1</sup>) while ML models predicted high values in the same LC. In Coata, SOCS predictions are similar for all LC types. As in Torobamba, the higher values were observed in the Tree (59.60 Mg ha<sup>-1</sup>) and Shrubland (57.49 Mg ha<sup>-1</sup>) areas, with the lowest values observed in Cropland (51.89 Mg ha<sup>-1</sup>).

#### 3.6. Validation and model performance

Table 5 shows the model performance metrics for each location. These metrics help assess the accuracy and reliability of the SOCS predictions made by the models. The data presents models with high accuracy and discusses their potential application in regions that share similar characteristics. The GWRCK was the best model when assessing

**Table 4**  
Summary of the coefficients used in geographically weighted regression model (GWR), and GWR with local collinearity adjustment (GWRC).

| Covariates                                   | GWR     |         |         | GWRC    |           |         |
|----------------------------------------------|---------|---------|---------|---------|-----------|---------|
|                                              | Minimum | Maximum | Median  | Minimum | Maximum   | Median  |
| Torobamba                                    |         |         |         |         |           |         |
| Intercept                                    | 50.65   | 54.13   | 52.45   | −19.41  | 109.8     | 45.95   |
| Second Brightness Index (BI2)                | −311.99 | −242.2  | −276.76 | −868.29 | 394.62    | −287.94 |
| Bare Soil Index (BSI)                        | −6.89   | 1.13    | −3.01   | −61.02  | 84.29     | −8.81   |
| Length Slope Factor (LS_factor)              | 0.16    | 0.24    | 0.19    | −0.3    | 0.61      | −0.01   |
| Normalized Height (N_height)                 | 22.51   | 27.64   | 25.41   | −18.53  | 73.35     | 33.92   |
| Plan Curvature (PLcurv)                      | −583.62 | 690.22  | 20.59   | −13,224 | 10,612.41 | −6880.9 |
| Profile Curvature (PRcurv)                   | −579.65 | 46.46   | −238.7  | −3959.2 | 1253.14   | −1576.5 |
| Soil Adjusted Total Vegetation Index (SATVI) | 135.94  | 139.61  | 137.67  | 9.72    | 281.42    | 93.77   |
| Topographic Position Index (TPI)             | 0.61    | 0.75    | 0.67    | −1.92   | 3.34      | 1.01    |
| Valley Depth (V_depth)                       | 0.02    | 0.03    | 0.03    | −0.06   | 0.12      | 0.04    |
| Coata                                        |         |         |         |         |           |         |
| Intercept                                    | 73.21   | 74.96   | 74.24   | 63.29   | 92.63     | 76.39   |
| Normalized Height (N_height)                 | −27.43  | −24.44  | −26.05  | −42.84  | −6.27     | −24.61  |
| Bare Soil Index (BSI)                        | −19.51  | −14.35  | −17.6   | −66.44  | −1.3      | −28.84  |
| Length Slope Factor (LS_factor)              | 0.93    | 0.97    | 0.95    | −0.02   | 0.79      | 0.56    |
| Plan Curvature (PLcurv)                      | −257.75 | −252.08 | −255.28 | −262.93 | −172.64   | −225.5  |
| Profile Curvature (PRcurv)                   | −3260.3 | −2556.6 | −2937.1 | −6246.7 | −499.83   | −3116.3 |
| Valley Depth (V_depth)                       | −0.06   | −0.05   | −0.05   | −0.1    | −0.03     | −0.05   |
| Soil Adjusted Total Vegetation Index (SATVI) | 44.66   | 52.15   | 49.38   | 5.44    | 72.84     | 50.02   |
| Topographic Position Index (TPI)             | 1.15    | 1.27    | 1.23    | −0.23   | 1.91      | 1.02    |
| Topographic Wetness Index (TWI)              | −1.34   | −1.29   | −1.31   | −1.76   | −0.97     | −1.33   |



**Fig. 3.** Spatial distribution of local  $R^2$  in a) Torobamba; b) Coata.

the datasets based on RMSE, MAE, and  $R^2$  for both study areas.

4. Discussion

4.1. Values of SOC and BD

Soils in Torobamba exhibited significantly higher SOC concentrations (2.52 %) compared to Coata (1.34 %). The SOC values in

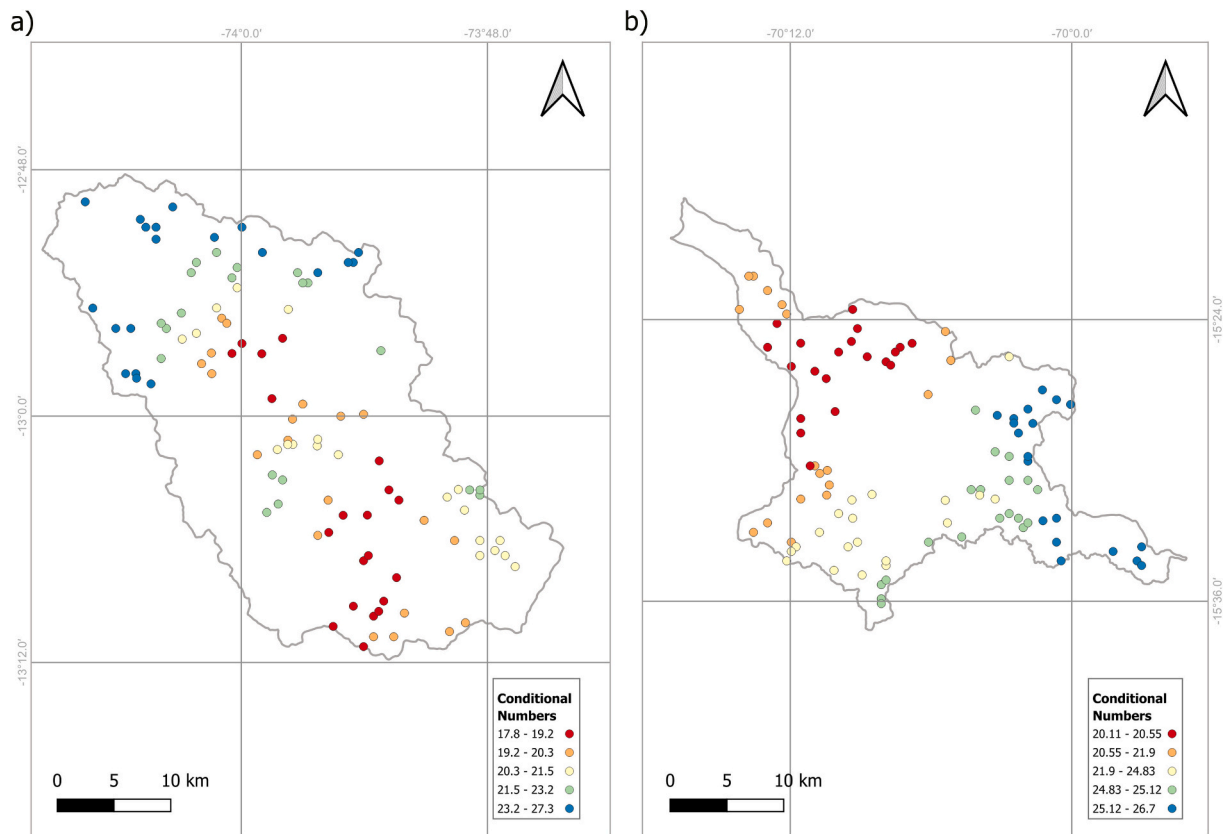


Fig. 4. Spatial distribution of local condition numbers (CN) values in a) Torobamba; b) Coata.

Torobamba surpassed those reported for Ferralsols, Acrisols, Planosols, and Podzols in the Amazon (2.18 %) (Castañeda-Martín and Montes-Pulido, 2017), yet remained lower than levels reported in cultivated pastures and grasslands in the Peruvian Andes (3.94 %, 3.83 %) (Carbajal et al., 2024). The BD values in this study (1.24–1.32  $\text{Mgm}^{-3}$ ) were notably higher than those reported for Andean soils under diverse land uses (0.48–0.53  $\text{gcm}^{-3}$ ) (Bravo-Medina et al., 2023). Similarly, BD values exceeded those documented in watersheds of the Ecuadorian and Colombian Andes (0.5 to 1.0  $\text{g cm}^{-3}$ ) (Andrade et al., 2022; Beltrán-Dávalos et al., 2022), suggesting higher levels of soil compaction. These differences likely reflect interactions among climate, vegetation, land use, and topography.

#### 4.2. Influence of explanatory variables on SOCS predictions with GWR and GWRC models

Geographically Weighted Regression (GWR) models confirmed spatial non-stationarity in the relationships between predictor variables and SOCS across the study areas (Table 4). In Torobamba, terrain curvature (PLcurv, PRCurv) and vegetation indices (particularly SATVI) dominated, while in Coata PR\_curv exerted the strongest influence, followed by N\_height and BSI. Coefficient ranges demonstrated strong spatial variability, often switching between positive and negative influences depending on location. Kernel function modifications and bandwidth parameters selection were found to directly affect GWR coefficient estimation (Wang et al., 2012).

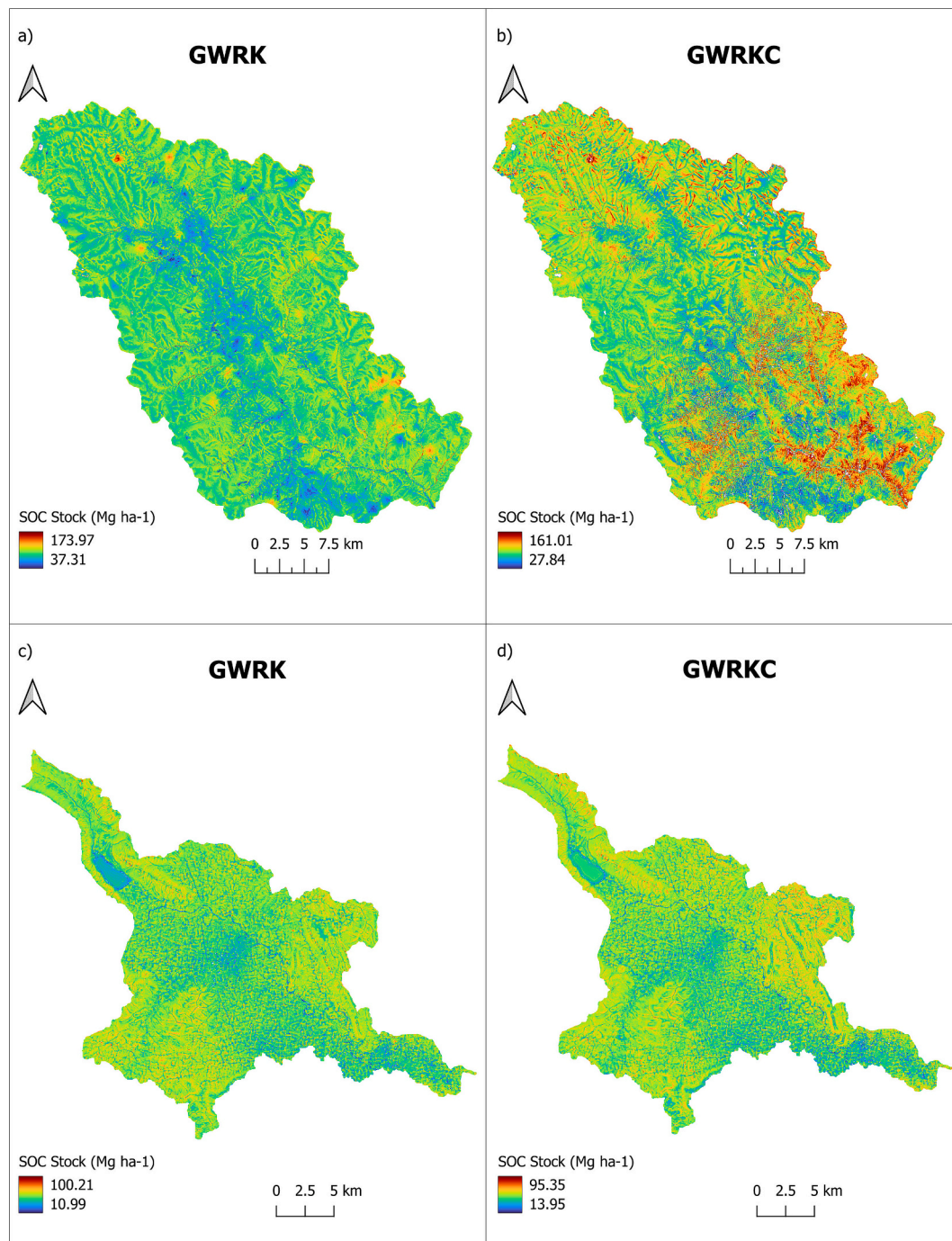
Comparisons between standard GWR and GWRC revealed key differences in coefficient distributions. GWRC produced broader coefficient ranges (e.g., PL\_curv in Torobamba: −583.62 to 690.22 in GWR vs. to −13,224.00 to 10,612.41 in GWRC). This underscores the effectiveness of GWRC in addressing local collinearity, improving interpretability of local regression coefficients despite minimal global collinearity (Wheeler, 2007). While GWRC enhances coefficient reliability under

multicollinearity (Bárcena et al., 2014), the increased variability requires careful interpretation. Bandwidth optimization was particularly critical, with adaptive bandwidths proving more effective under uneven sample densities (Kiani et al., 2024).

Despite accounting for spatial variation, local  $R^2$  values remained low (mean  $R^2$ : 0.136 in Torobamba, 0.177 in Coata) (Fig. 3), suggesting selected predictors only partially explained SOCS variability. Possible causes include missing explanatory variables, unaddressed non-linearities, or inherent system randomness. Strong spatial heterogeneity, unrecorded local effects, or locations where the underlying spatial process is inherently noisy or less predictable likely contribute to these results. In regions with complex geomorphology and spatial heterogeneity, finding a clear pattern can be difficult (Costa et al., 2018). Sparse sampling further limited local reliability (Devkota et al., 2014) while residual spatial autocorrelation pointed to model limitations (Gaspard et al., 2019). Bandwidth sensitivity further influenced outcomes: overly broad bandwidths risked over-smoothing and convergence toward a standard regression model (Weku et al., 2022), which can lead to. Multicollinearity (CN values ~20–28) (Fig. 4) can produce unstable coefficient estimates highly sensitive to minor changes, impacting the interpretation of local predictor effects (Brunsdon et al., 2012).

GWR outperformed OLS by reducing AICc (Sartika and Suryani, 2020), explicitly capturing spatial heterogeneity and local variation often ignored by conventional ML models (Yang et al., 2022; Mishra et al., 2010). GWRK improved further by combining local regression with kriging of residuals (Kumar et al., 2012; Imran et al., 2015). While standalone ML offers predictive accuracy, it lacks spatial interpretability and risks bias without adaptations such as Random Forest Spatial Interpolation (Sun et al., 2021; Kmoch et al., 2025). Integration of GWR and kriging remains a robust approach for SOCS mapping (Adeniyi et al., 2024; Faisal et al., 2025).

GWRC provided more stable and interpretable local coefficient by mitigating multicollinearity effects (Czarnota et al., 2015), though at the



**Fig. 5.** Spatial distribution of SOCS predicted values in a) Torobamba using Geographically Weighted Regression Kriging (GWRK); b) Torobamba employing Geographically Weighted Regression with local collinearity adjustment Kriging (GWRCK); c) Coata using GWRK; and d) Coata employing GWRCK.

expense of wider coefficient ranges or altered weighting schemes. Even though these changes are designed to address collinearity, local models may still use different neighbor sets or weights, which may account for the differences in the GWR coefficient estimates (Leong and Yue, 2017). Multi-scale GWR (MGWR) may further improve performance by capturing spatial heterogeneity and residual structure more effectively than OLS and supporting accurate, spatially explicit carbon stock estimates (Song et al., 2024; Kuang and Chen, 2025).

#### 4.3. Variable importance (VI) in machine learning models

Analyzing variable importance provides crucial insights into model

functionality and is vital for improving performance (Chamma et al., 2024). High VI scores indicate strong predictive power and a close relationship between a predictor and SOC variation in the study area. However, establishing causality is different, as it requires evaluating whether randomizing the predictor reduces prediction accuracy (Lamsaf et al., 2025; Wang et al., 2024). Numerous studies have shown topographic and vegetation variables to be critical predictors of SOC. Topography can directly influence SOC accretion in valley bottoms (Schwanghart and Jarmer, 2011), or indirectly reflect soil moisture conditions regulating vegetation growth and residue decomposition (Rasel et al., 2017). In the present study,  $V_{\text{depth}}$  strongly influenced both basins (Fig. 4). Topographic variables derived from DEMs, such as

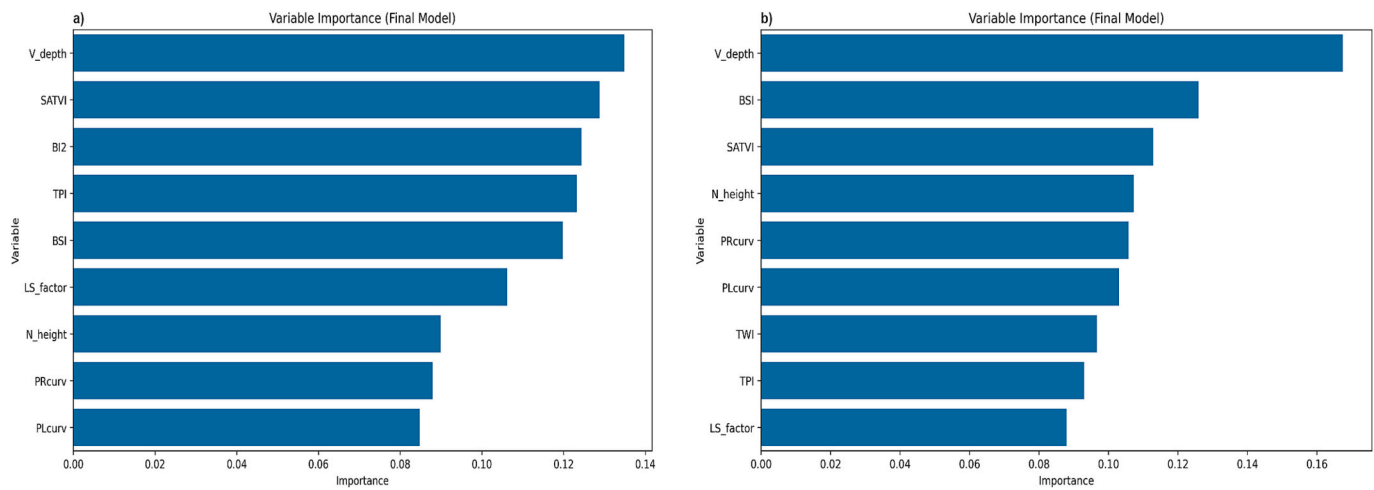


Fig. 6. Feature of importance plots for the Random Forest model in a) Torobamba; b) Coata.

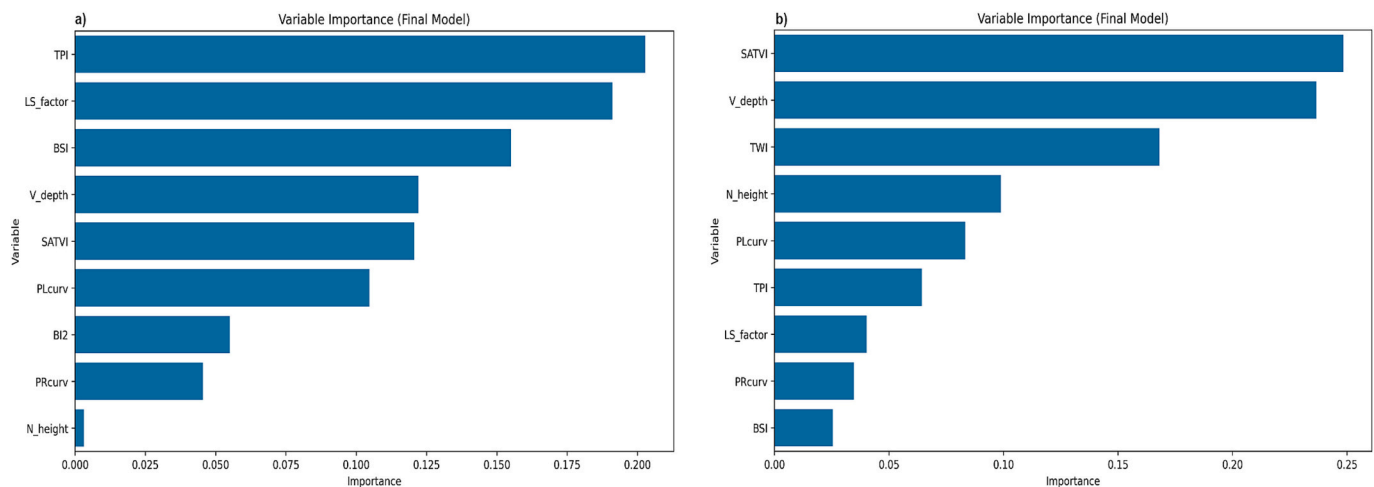


Fig. 7. Feature of importance plots for the Gradient Boosting model in a) Torobamba; b) Coata.

TWI and V\_depth, affect SOC erosion and deposition processes (Dorji et al., 2014; Wang et al., 2020). Previous research found V\_depth to have moderate influence in SOC stock prediction compared to elevation and LS\_factor (Adhikari et al., 2019), while other models such as RF and BRT identified standardized height and altitude as best predictors (Ließ et al., 2016). In Coata, the relative homogeneous terrain limited topographic influence, with TWI emerging as the primary variable (John et al., 2020).

Land use/land cover and vegetation indices are also critical SOC predictors in the Peruvian Andes (Carbajal et al., 2024). Precipitation and NDVI have been found to outperform terrain attributes (Emadi et al., 2020). In this study, SATVI was a crucial predictor in Torobamba, while BSI was more important in Coata. BSI combined with alongside geological units, soil taxonomy, elevation, and the LS\_factor, has improved SOC mapping accuracy (Ayala Izurieta et al., 2021). Recent findings show that combining spectral indices with topographical features reduces validation errors and optimizes SOC predictions (Cutting et al., 2024).

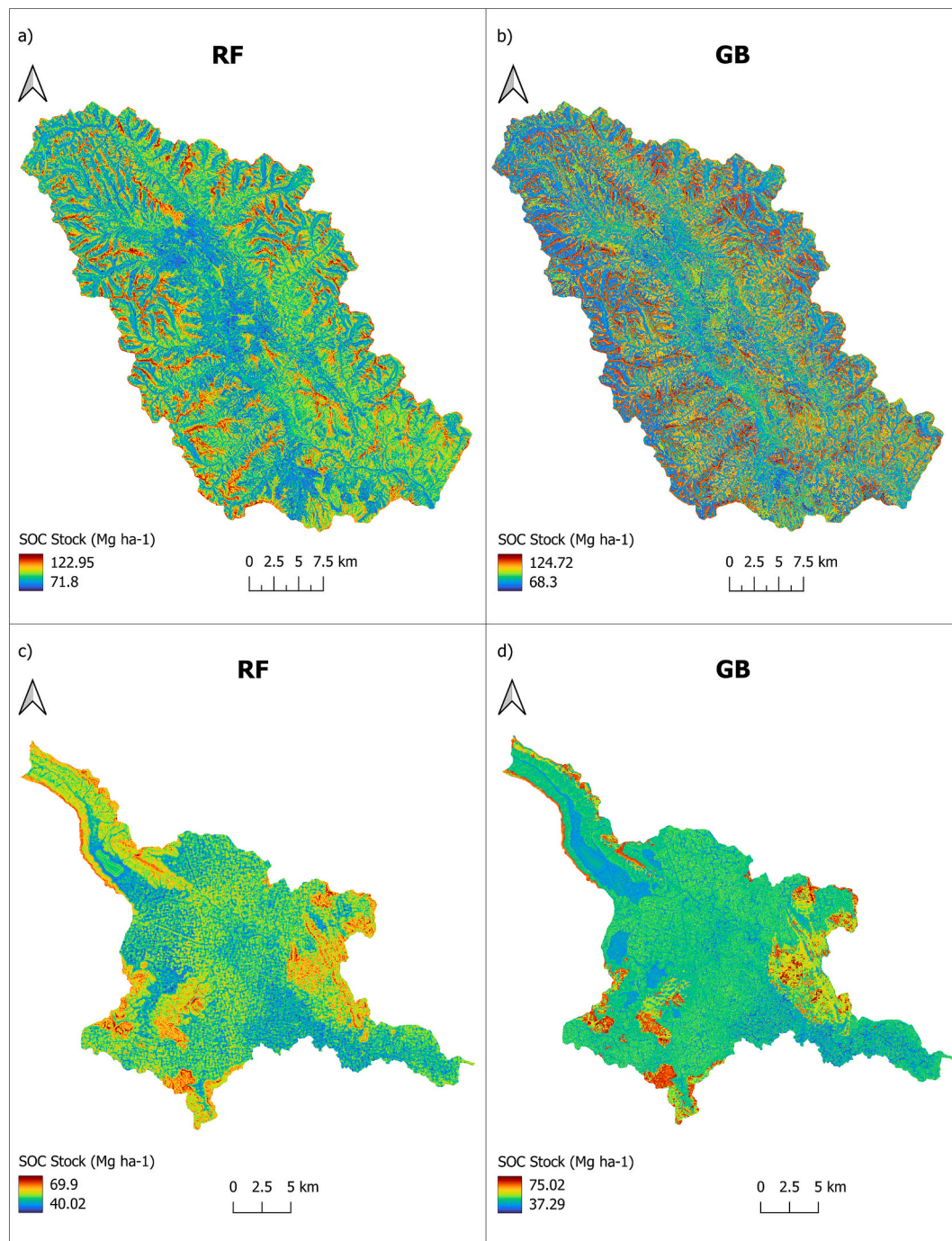
#### 4.4. SOCS prediction using GWR and ML models

All four modeling approaches generated similar within-basin predictions, though inter-basin differences were substantial. In Torobamba, GWRK predicted the highest SOCS ( $173.97 \text{ Mg ha}^{-1}$ ), followed by GWRKC ( $161.01 \text{ Mg ha}^{-1}$ ), while GB estimates were much lower ( $68.3$

$\text{Mg ha}^{-1}$ ). In Coata, GWRK also produced the highest SOCS ( $100.21 \text{ Mg ha}^{-1}$ ), while GB again predicted the lowest ( $37.29 \text{ Mg ha}^{-1}$ ). These differences underscore the need for ensemble methods when estimating SOCS (Mishra et al., 2020). While GWR models proved useful in areas of high spatial SOCS variability, ML produced narrower SOCS ranges, failing to fully capture the inherent variability, potentially due to sampling limitations.

#### 4.5. SOCS under different land cover types

Distinct SOCS patterns emerged across land cover types in both basins. In Torobamba (Fig. 9), shrublands showed the highest SOC stocks (mean:  $94.93 \text{ Mg ha}^{-1}$ ), followed by grasslands ( $91.60 \text{ Mg ha}^{-1}$ ), tree cover ( $91.60 \text{ Mg ha}^{-1}$ ), and croplands ( $88.95 \text{ Mg ha}^{-1}$ ). The SOCS in shrublands may be attributed to the combination of slower decomposition rates and substantial biomass input from shrub species, which contribute to organic matter accumulation in the soil (Liu et al., 2023). In Coata (Fig. 10), tree cover exhibited the highest SOCS (mean:  $56.71 \text{ Mg ha}^{-1}$ ), followed by shrublands ( $56.32 \text{ Mg ha}^{-1}$ ), with grasslands ( $54.42 \text{ Mg ha}^{-1}$ ) and croplands ( $52.22 \text{ Mg ha}^{-1}$ ) showing lower values. These results align with global findings that forests typically maintain higher SOCS due to continuous litter input and reduced soil disturbance compared to managed agricultural lands (Lal, 2004). The consistently lower cropland SOCS across both basins reflects the impact of agricultural activities on soil carbon dynamics. These findings corroborate that



**Fig. 8.** Maps of predicted values of Soil Organic Carbon Stocks (SOCs) in a) Torobamba using Random Forest (RF), b) Torobamba using Gradient Boosting (GB), c) Coata using RF, and d) Coata using GB.

significant SOC losses occur after converting natural ecosystems to cropland due to reduced carbon input, increased decomposition rates, and soil disturbance from tillage practices (Deng et al., 2016; Wei et al., 2014).

Environmental differences, such as precipitation and temperature, may partly explain greater biomass production and slower decomposition rates in Torobamba, contributing to higher SOCS relative to Coata. Such regional variations in SOCS have been reported, emphasizing the importance of climate gradients and geomorphological features in determining regional SOC distribution patterns (Wiesmeier et al., 2019). Despite numerical differences among models, consistency in relative patterns among land cover types increases confidence in the overall

findings.

#### 4.6. Performance of models for predicting and mapping SOCS

GWR and its variants, RF, and GB models to reveal significant differences in performance to predict SOCS under Andean conditions (Table 5). In Torobamba, GWR variants (GWRK and GWRCK) achieved the highest predictive performance, with RMSE reductions of 79–90 % and  $R^2$  values up to 0.99. Kriging interpolation enhanced predictions by addressing residual spatial autocorrelation. These findings are consistent with prior research indicating the effectiveness of GWR models in addressing spatial non-stationarity (Cao et al., 2022; Li et al., 2010). This

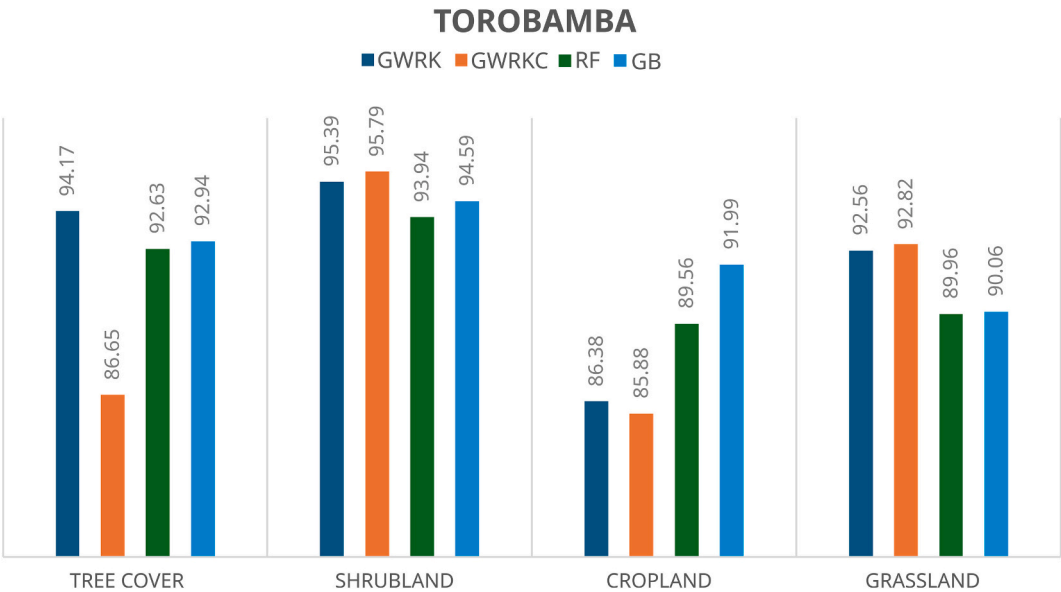


Fig. 9. Distribution of predicted Soil Organic Carbon Stocks in Mg ha<sup>-1</sup> on Land Cover Types using all models in Torobamba.

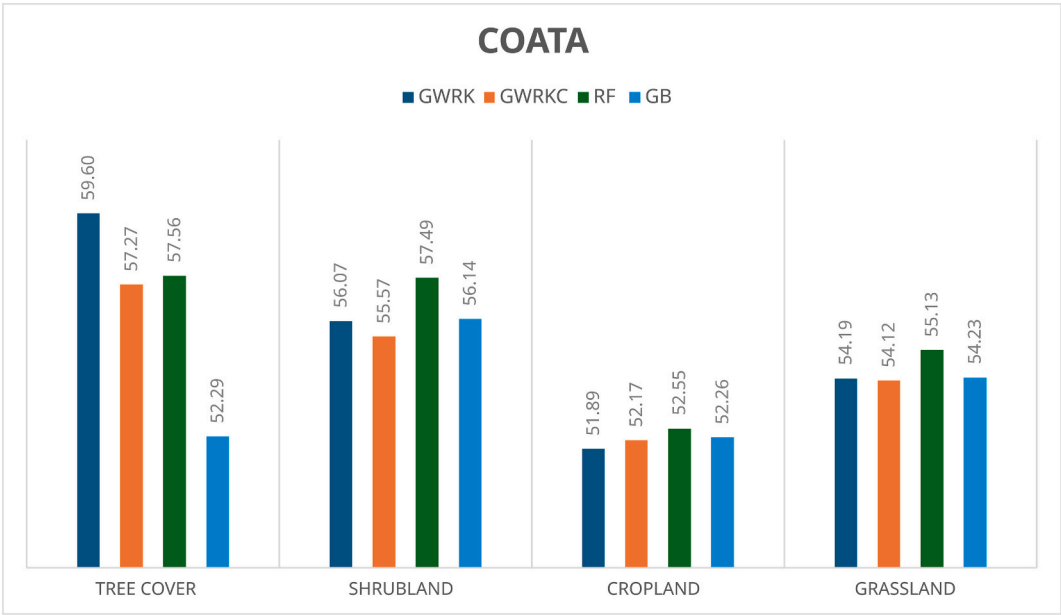


Fig. 10. Distribution of predicted Soil Organic Carbon Stocks in Mg ha<sup>-1</sup> on Land Cover Types using all models in Coata.

**Table 5**  
Summary of performance metrics for Geographically Weighted Regression (GWR) and their kriging adjusted (GWRK), GWR with collinearity analysis (GWRC) and their kriging adjusted (GWRCK), Random Forest (RF), and Gradient Boosting (GB) models.

| Basin     | Metrics        | GWR   | GWRK  | Change <sup>1</sup> (%) | GWRC  | GWRCK | Change <sup>1</sup> (%) | RF    | GB    |
|-----------|----------------|-------|-------|-------------------------|-------|-------|-------------------------|-------|-------|
| Torobamba | RMSE           | 31.28 | 6.52  | 79.14                   | 25.26 | 2.48  | 90.17                   | 29.29 | 30.67 |
|           | MAE            | 26.76 | 5.57  | 79.18                   | 20.70 | 2.02  | 90.22                   | 26.11 | 27.57 |
|           | R <sup>2</sup> | 0.14  | 0.96  | 609.36                  | 0.44  | 0.99  | 127.77                  | 0.11  | 0.03  |
| Coata     | RMSE           | 15.60 | 14.37 | 7.84                    | 14.89 | 13.43 | 9.80                    | 14.19 | 16.01 |
|           | MAE            | 12.74 | 11.74 | 7.88                    | 12.21 | 11.01 | 9.87                    | 13.15 | 14.75 |
|           | R <sup>2</sup> | 0.17  | 0.30  | 71.48                   | 0.25  | 0.39  | 56.81                   | -0.01 | -0.28 |

<sup>1</sup> Percentage improvement observed with the application of Kriging Interpolation.

attribute is critical in heterogeneous Andean landscapes with variable topography and microclimates. In Coata, improvements were smaller (RMSE reduction 7.8–9.8 %) though R<sup>2</sup> values (0.3–0.39) still

outperformed other models, suggesting less pronounced spatial variability or limitations in capturing local effects. RF and GB performed poorly in both basins. In Torobamba, R<sup>2</sup> values

remained below 0.11, while in Coata, RF (RMSE 14.19) and GB (RMSE 16.01) showed moderate performance but still produced negative  $R^2$  values, suggesting poor fit. These outcomes may stem from inadequate feature selection, suboptimal hyperparameter tuning, insufficient training data, or lack of spatial covariates. While RF often excels in SOC prediction (Hengl et al., 2018) (Ayala Izurieta et al., 2021; Ließ et al., 2016), its performance here indicates challenges unique to Andean conditions. One reason might be that non-spatial ML models may not adequately capture the complex spatial dependencies in complex landscapes. Another one might be that the model performance is highly context dependent. In contrast to the relatively uniform landscape of Coata, the rugged terrain of Torobamba may enhance the usefulness of GWR. Conversely, the environmental and land-use patterns in Coata could either mask spatial variations or introduce confounding effects.

Several studies have shown that ML and GWR models developed for SOCS prediction in one region may be transferable to others when similarities in climate, soil types, land use, and covariate datasets exist. However, differences in local environmental drivers and data availability often require model recalibration or local adaptation. Cross-regional transfer is most effective when models rely on harmonized datasets, such as the Harmonized World Soil Database (HWSD) or WISE30se (Batjes, 2016). Recent advances, such as domain-adaptive ML for crop yield prediction (Priyatikanto et al., 2023), pasture growth transfer modeling across diverse climates to predict nitrogen response rates (Pylaniadis et al., 2024), and SOC prediction using Convolutional Neural Network-based transfer learning- (Han et al., 2025) highlight promising future directions.

## 5. Conclusions

Digital SOCS mapping provides critical insights for characterizing high Andean soils in Peru, supporting the development of regulatory frameworks to prevent degradation and contributing to global SOC assessments. These soils play critical roles in regulating hydrological processes, sequestering atmospheric carbon and providing ecosystem services, underscoring their regional and global significance. The high SOCS values obtained in the present study highlight the importance of characterizing SOC reserves in Andean landscapes.

Our results demonstrated that GWR and its variants outperformed conventional ML for SOCS prediction in complex Andean landscapes, especially under small to medium-sized sampling GWRK, in particular, accurately captured spatial heterogeneity, enhancing predictive reliability. By contrast, RF and GB showed limited performance without explicit spatial adaptation mechanisms.

Although SOCS modeling techniques such as GWR and ML have broad applicability, they require thorough validation and context-specific adjustments to ensure reliable predictions. Future research should explore hybrid modeling frameworks that integrate spatial weighting and ML to optimize predictions in heterogeneous mountain environments.

## CRedit authorship contribution statement

**Carlos Carbajal:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Merely Tumbalobos-Dextre:** Visualization, Software, Methodology. **Tatiana Condori-Ataupillco:** Resources, Investigation. **Nestor Cuellar-Condori:** Resources, Investigation. **Carla Gavilan:** Writing – review & editing, Conceptualization.

## Funding

This research was funded by the INIA project “Mejoramiento de los servicios de investigación y transferencia tecnológica en el manejo y recuperación de suelos agrícolas degradados y aguas para riego en la pequeña y mediana agricultura en los departamentos de Lima, Áncash,

San Martín, Cajamarca, Lambayeque, Junín, Ayacucho, Arequipa, Puno y Ucayali”, CUI 2487112.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgments

To the Soil, Water, and Foliar Laboratory (LABSAF) network technicians, especially of La Molina, Canaán, and Illpa Experimental Agrarian Stations headquarters. Special thanks go to Marilia Coila Mamani and Fredy Flores Galindo for their help collecting soil samples.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.geodrs.2025.e01026>.

## Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

## References

- Adeniyi, O.D., Brenning, A., Maerker, M., 2024. Spatial prediction of soil organic carbon: combining machine learning with residual kriging in an agricultural lowland area (Lombardy region, Italy). *Geoderma* 448, 116953. <https://doi.org/10.1016/j.geoderma.2024.116953>.
- Adhikari, K., Owens, P.R., Libohova, Z., Miller, D.M., Wills, S.A., Nemecek, J., 2019. Assessing soil organic carbon stock of Wisconsin, USA and its fate under future land use and climate change. *Sci. Total Environ.* 667, 833–845. <https://doi.org/10.1016/j.scitotenv.2019.02.420>.
- Anderson, E., Marengo, J., Villalba, R., Halloy, S., Young, B., Cordero, D., Gast, F., Jaimes, E., Ruiz Carrascal, D., 2011. Consequences of climate change for ecosystems and ecosystem services in the tropical Andes. In: *Climate Change and Biodiversity in the Tropical Andes*. MacArthur Foundation Inter American Institute For Global Change Research. IAI The Scientific Committee on Problems of the Environment. SCOPE, pp. 1–18.
- Andrade, H.J., Segura, M.A., Canal-Daza, D.S., 2022. Conservation of soil organic carbon in the National Park Santuario de Fauna y Flora Iguaque. *Boyacá-Colombia For.* 13, 1275. <https://doi.org/10.3390/f13081275>.
- Ayala Izurieta, J.E., Márquez, C.O., García, V.J., Jara Santillán, C.A., Sisti, J.M., Pasqualotto, N., Van Wittenberghe, S., Delegado, J., 2021. Multi-predictor mapping of soil organic carbon in the alpine tundra: a case study for the central Ecuadorian Páramo. *Carbon Balance Manag.* 16, 32. <https://doi.org/10.1186/s13021-021-00195-2>.
- Bárcena, M.J., Menéndez, P., Palacios, M.B., Tusell, F., 2014. Alleviating the effect of collinearity in geographically weighted regression. *J. Geogr. Syst.* 16, 441–466. <https://doi.org/10.1007/s10109-014-0199-6>.
- Baret, F., Guyot, G., 1991. Potentials and limits of vegetation indices for LAI and APAR assessment. *Remote Sens. Environ.* 35, 161–173. [https://doi.org/10.1016/0034-4257\(91\)90009-U](https://doi.org/10.1016/0034-4257(91)90009-U).
- Batjes, N.H., 2016. Harmonized soil property values for broad-scale modelling (WISE30sec) with estimates of global soil carbon stocks. *Geoderma* 269, 61–68. <https://doi.org/10.1016/j.geoderma.2016.01.034>.
- Beltrán-Dávalos, A.A., Ayala Izurieta, J.E., Echeverría Guadalupe, M.M., Van Wittenberghe, S., Delegado, J., Otero Pérez, X.L., Merino, A., 2022. Evaluation of soil organic carbon storage of Attilo in the Ecuadorian Andean wetlands. *Soil Syst.* 6, 92. <https://doi.org/10.3390/soilsystems6040092>.
- Belyadi, H., Haghighat, A., 2021. Chapter 5 - supervised learning. In: Belyadi, H., Haghighat, A. (Eds.), *Machine Learning Guide for Oil and Gas Using Python*. Gulf Professional Publishing, pp. 169–295. <https://doi.org/10.1016/B978-0-12-821929-4.00004-4>.
- Bivand, R., Yu, D., 2006. spgwr: Geographically Weighted Regression. <https://doi.org/10.32614/CRAN.package.spgwr>.
- Bravo-Medina, C., Torres-Navarrete, B., Arteaga-Crespo, Y., García-Quintana, Y., Reyes-Morán, H., Changoluisa-Vargas, D., Paguay-Sayay, D., 2023. Soil properties variation in a small-scale altitudinal gradient of an evergreen foothills forest, Ecuadorian Amazon region. *Eur. J. For. Res.* 142, 1325–1339. <https://doi.org/10.1007/s10342-023-01593-6>.
- Breiman, L., 2001. Random forests. *Mach. Learn.* 45, 5–32. <https://doi.org/10.1023/A:1010933404324>.

- Brunsdon, C., Fotheringham, A.S., Charlton, M.E., 1996. Geographically weighted regression: a method for exploring spatial nonstationarity. *Geogr. Anal.* 28, 281–298. <https://doi.org/10.1111/j.1538-4632.1996.tb00936.x>.
- Brunsdon, C., Charlton, M., Harris, P., 2012. Living with Collinearity in Local Regression Models. *Spatial Accuracy*.
- Cao, Z., Zhu, W., Luo, P., Wang, S., Tang, Z., Zhang, Y., Guo, B., 2022. Spatially non-stationary relationships between changing environment and water yield Services in Watersheds of China's climate transition zones. *Remote Sens* 14, 5078. <https://doi.org/10.3390/rs14205078>.
- Carbajal, M., Ramírez, D.A., Turin, C., Schaeffer, S.M., Konkell, J., Ninanya, J., Rinza, J., De Mendiburu, F., Zorogastua, P., Villaorduña, L., Quiroz, R., 2024. From rangelands to cropland, land-use change and its impact on soil organic carbon variables in a peruvian andean highlands: a machine learning modeling approach. *Ecosystems*. <https://doi.org/10.1007/s10021-024-00928-7>.
- Castañeda-Martín, A.E., Montes-Pulido, C.R., 2017. Carbono almacenado en páramo andino. *Entomado* 13, 210–221. <https://doi.org/10.18041/entomado.2017v13n1.25112>.
- Chamma, A., Thirion, B., Engemann, D., 2024. Variable importance in high-dimensional settings requires grouping. *AAAI* 38, 11195–11203. <https://doi.org/10.1609/aaai.v38i10.28997>.
- Chen, J., Qu, M., Zhang, J., Xie, E., Zhao, Y., Huang, B., 2021. Improving the spatial prediction accuracy of soil alkaline hydrolyzable nitrogen using GWPCA-GWRK. *Soil Sci. Soc. Am. J.* 85, 879–892. <https://doi.org/10.1002/saj2.20189>.
- Chen, S., Arrouays, D., Mulder, V.L., Poggio, L., Minasny, B., Roudier, P., Libohova, Z., Lagacherie, P., Shi, Z., Hannam, J., Meersmans, J., Richer-de-Forges, A.C., Walter, C., 2022. Digital mapping of GlobalSoilMap soil properties at a broad scale: a review. *Geoderma* 409, 115567. <https://doi.org/10.1016/j.geoderma.2021.115567>.
- Chen, Q., Wang, Y., Zhu, X., 2024. Soil organic carbon estimation using remote sensing data-driven machine learning. *PeerJ* 12, e17836. <https://doi.org/10.7717/peerj.17836>.
- Conrad, O., Bechtel, B., Bock, M., Dietrich, H., Fischer, E., Gerlitz, L., Wehberg, J., Wichmann, V., Böhner, J., 2015. System for automated geoscientific analyses (SAGA) v. 2.1.4. *Geosci. Model Dev.* 8, 1991–2007. <https://doi.org/10.5194/gmd-8-1991-2015>.
- Costa, E.M., de Tassinari, W.S., Pinheiro, H.S.K., Beutler, S.J., dos Anjos, L.H.C., 2018. Mapping soil organic carbon and organic matter fractions by geographically weighted regression. *J. Environ. Qual.* 47, 718–725. <https://doi.org/10.2134/jeq2017.04.0178>.
- Cutting, B.J., Atzberger, C., Gholizadeh, A., Robinson, D.A., Mendoza-Ulloa, J., Marti-Cardona, B., 2024. Remote quantification of soil organic carbon: role of topography in the intra-field distribution. *Remote Sens* 16, 1510. <https://doi.org/10.3390/rs16091510>.
- Czarnota, J., Wheeler, D.C., Gennings, C., 2015. Evaluating geographically weighted regression models for environmental chemical risk analysis. *Cancer Informat.* 14, 117–127. <https://doi.org/10.4137/CIN.S17296>.
- Deng, L., Zhu, G., Tang, Z., Shanguan, Z., 2016. Global patterns of the effects of land-use changes on soil carbon stocks. *Glob. Ecol. Conserv.* 5, 127–138. <https://doi.org/10.1016/j.gecco.2015.12.004>.
- Devkota, M., Hatfield, G., Chintala, R., 2014. Effect of sample size on the performance of ordinary least squares and geographically weighted regression. *Br. J. Math. Comput. Sci.* 4, 1–21. <https://doi.org/10.9734/BJMCS/2014/6050>.
- Dorji, T., Odeh, I.O.A., Field, D.J., Baillie, I.C., 2014. Digital soil mapping of soil organic carbon stocks under different land use and land cover types in montane ecosystems, eastern Himalayas. *For. Ecol. Manag.* 318, 91–102. <https://doi.org/10.1016/j.foreco.2014.01.003>.
- Emadi, M., Taghizadeh-Mehrjardi, R., Cherati, A., Danesh, M., Mosavi, A., Scholten, T., 2020. Predicting and mapping of soil organic carbon using machine learning algorithms in northern Iran. *Remote Sens* 12, 2234. <https://doi.org/10.3390/rs121242234>.
- Emamgholizadeh, S., Shahsavani, S., Eslami, M.A., 2017. Comparison of artificial neural networks, geographically weighted regression and Cokriging methods for predicting the spatial distribution of soil macronutrients (N, P, and K). *Chin. Geogr. Sci.* 27, 747–759. <https://doi.org/10.1007/s11769-017-0906-6>.
- Escadafal, R., 1989. Remote sensing of arid soil surface color with Landsat thematic mapper. *Adv. Space Res.* 9, 159–163. [https://doi.org/10.1016/0273-1177\(89\)90481-X](https://doi.org/10.1016/0273-1177(89)90481-X).
- Faisal, F., Pramoddyo, H., Astutik, S., Efendi, A., 2025. Bayesian geographically weighted regression with kriging for enhanced spatial prediction: a comparison of Jeffreys' and conjugate priors. *Math. Model. Eng. Probl.* 12. <https://doi.org/10.18280/mmep.120515>.
- Fick, S.E., Hijmans, R.J., 2017. WorldClim 2: new 1-km spatial resolution climate surfaces for global land areas. *Int. J. Climatol.* 37, 4302–4315. <https://doi.org/10.1002/joc.5086>.
- Fisher, A., Rudin, C., Dominici, F., 2019. All models are wrong, but many are useful: learning a variable's importance by studying an entire class of prediction models simultaneously. *J. Mach. Learn. Res.* 20, 1–81.
- Fotheringham, A.S., Charlton, M.E., Brunsdon, C., 1998. Geographically weighted regression: a natural evolution of the expansion method for spatial data analysis. *Environ. Plan. A* 30, 1905–1927. <https://doi.org/10.1068/a301905>.
- Friedman, J.H., 2001. Greedy function approximation: a gradient boosting machine. *Ann. Stat.* 29, 1189–1232. <https://doi.org/10.1214/aos/1013203451>.
- Gao, B., 1996. NDWI—A normalized difference water index for remote sensing of vegetation liquid water from space. *Remote Sens. Environ.* 58, 257–266. [https://doi.org/10.1016/S0034-4257\(96\)00067-3](https://doi.org/10.1016/S0034-4257(96)00067-3).
- García Lino, M.C., Pfanzelt, S., Domic, A.I., Hensen, I., Schitteck, K., Meneses, R.I., Bader, M.Y., 2024. Carbon dynamics in high-Andean tropical cushion peatlands: a review of geographic patterns and potential drivers. *Ecol. Monogr.* 94, e1614. <https://doi.org/10.1002/ecm.1614>.
- Gaspard, G., Kim, D., Chun, Y., 2019. Residual spatial autocorrelation in macroecological and biogeographical modeling: a review. *J. Ecol. Environ.* 43, 19. <https://doi.org/10.1186/s41610-019-0118-3>.
- Genier, R., Poggi, J.-M., Tuleau-Malot, C., Villa-Vialaneix, N., 2017. Random forests for big data. *Big Data Res.* 9, 28–46. <https://doi.org/10.1016/j.bdr.2017.07.003>.
- Ghaderpour, E., Mazzanti, P., Bozzano, F., Scarascia Mugnozza, G., 2024. Trend analysis of MODIS land surface temperature and land cover in Central Italy. *Land* 13, 796. <https://doi.org/10.3390/land13060796>.
- Gitelson, A.A., Kaufman, Y.J., Merzlyak, M.N., 1996. Use of a green channel in remote sensing of global vegetation from EOS-MODIS. *Remote Sens. Environ.* 58, 289–298. [https://doi.org/10.1016/S0034-4257\(96\)00072-7](https://doi.org/10.1016/S0034-4257(96)00072-7).
- Gollini, I., Lu, B., Charlton, M., Brunsdon, C., Harris, P., 2015. GWmodel: an R package for exploring spatial heterogeneity using geographically weighted models. *J. Stat. Softw.* 63. <https://doi.org/10.18637/jss.v063.i17>.
- Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore, R., 2017. Google earth engine: planetary-scale geospatial analysis for everyone. *Remote Sens. Environ.* 202, 18–27. <https://doi.org/10.1016/j.rse.2017.06.031>. Big Remotely Sensed Data: tools, applications and experiences.
- Guo, L., Luo, M., Zhangyang, C., Zeng, C., Wang, S., Zhang, H., 2018. Spatial modelling of soil organic carbon stocks with combined principal component analysis and geographically weighted regression. *J. Agric. Sci.* 156, 774–784. <https://doi.org/10.1017/S0021859618000709>.
- Habib, M., Habib, A., Alibrahim, B., 2024. Prediction and parametric assessment of soil one-dimensional vertical free swelling potential using ensemble machine learning models. *Adv. Model. Simul. Eng. Sci.* 11, 26. <https://doi.org/10.1186/s40323-024-00277-z>.
- Halder, K., Srivastava, A.K., Ghosh, A., Nabik, R., Pan, S., Chatterjee, U., Bisai, D., Pal, S. C., Zeng, W., Ewert, F., Gaiser, T., Pande, C.B., Islam, A.R.Md.T., Alam, E., Islam, M. K., 2024. Application of bagging and boosting ensemble machine learning techniques for groundwater potential mapping in a drought-prone agriculture region of eastern India. *Environ. Sci. Eur.* 36, 155. <https://doi.org/10.1186/s12302-024-00981-y>.
- Han, J., Wu, M., Qi, Y., Li, X., Chen, X., Wang, J., Zhu, J., Li, Q., 2025. A soil organic carbon mapping method based on transfer learning without the use of exogenous data. *Front. Environ. Sci.* 13, 1580085. <https://doi.org/10.3389/fenvs.2025.1580085>.
- Hengl, T., Nussbaum, M., Wright, M.N., Heuvelink, G.B.M., Gräler, B., 2018. Random forest as a generic framework for predictive modeling of spatial and spatio-temporal variables. *PeerJ* 6, e5518. <https://doi.org/10.7717/peerj.5518>.
- Hengl, T., Sorenson, P., Parente, L., Cornish, K., Battigelli, J., Bonannella, C., Gorzelak, M., Nichols, K., 2023. Assessment of soil organic carbon stocks in Alberta using 2-scale sampling and 3D predictive soil mapping. *FACETS* 8, 1–17. <https://doi.org/10.1139/facets-2023-0040>.
- Hiemstra, P.H., Pebesma, E.J., Twenhöfel, C.J.W., Heuvelink, G.B.M., 2009. Real-time automatic interpolation of ambient gamma dose rates from the Dutch radioactivity monitoring network. *Comput. Geosci.* 35, 1711–1721. <https://doi.org/10.1016/j.cageo.2008.10.011>.
- Hijmans, R.J., Barbosa, M., Ghosh, A., Mandel, A., 2021. geodata: Download Geographic Data. <https://doi.org/10.32614/CRAN.package.geodata>.
- Hollister, J., Shah, T., Nowosad, J., Robitaille, A.L., Beck, M.W., Johnson, M., 2023. Elevatr: Access Elevation Data from Various APIs (Manual). <https://doi.org/10.5281/zenodo.8335450>.
- Hounkpatin, K.O.L., Stendahl, J., Lundblad, M., Karlun, E., 2021. Predicting the spatial distribution of soil organic carbon stock in Swedish forests using a group of covariates and site-specific data. *SOIL* 7, 377–398. <https://doi.org/10.5194/soil-7-377-2021>.
- Huete, A.R., 1988. A soil-adjusted vegetation index (SAVI). *Remote Sens. Environ.* 25, 295–309. [https://doi.org/10.1016/0034-4257\(88\)90106-X](https://doi.org/10.1016/0034-4257(88)90106-X).
- Huete, A., Didan, K., Miura, T., Rodriguez, E.P., Gao, X., Ferreira, L.G., 2002. Overview of the radiometric and biophysical performance of the MODIS vegetation indices. *Remote Sens. Environ.* 83, 195–213. [https://doi.org/10.1016/S0034-4257\(02\)00096-2](https://doi.org/10.1016/S0034-4257(02)00096-2). The Moderate Resolution Imaging Spectroradiometer (MODIS): a new generation of Land Surface Monitoring.
- Imran, M., Stein, A., Zurita-Milla, R., 2015. Using geographically weighted regression kriging for crop yield mapping in West Africa. *Int. J. Geogr. Inf. Sci.* 29, 234–257. <https://doi.org/10.1080/13658816.2014.959522>.
- ISO, 1996. Soil quality - Determination of organic and total carbon after dry combustion (Standard No. ISO 10694:1996).
- ISO, 2017. Soil quality - Determination of dry bulk density (Standard No. ISO 11272: 2017).
- IUSS Working Group, 2007. World Reference Base for Soil Resources 2006, First Update 2007, World Soil Resources Reports no. 103. ed. FAO, Rome.
- Jenny, H., 1994. Factors of Soil Formation: A System of Quantitative Pedology, Unabridged, Unaltered Republ., New Foreword. Ed, Dover Books on Earth Sciences. Dover Publ, New York.
- John, K., Abraham Isong, I., Michael Kebonye, N., Okon Ayito, E., Chapman Agyeman, P., Marcus Afu, S., 2020. Using Machine Learning Algorithms to Estimate Soil Organic Carbon Variability with Environmental Variables and Soil Nutrient Indicators in an Alluvial Soil. *Land* 9, 487. <https://doi.org/10.3390/land9120487>.
- Kiani, B., Sartorius, B., Lau, C.L., Bergquist, R., 2024. Mastering geographically weighted regression: key considerations for building a robust model. *Geospat. Health* 19. <https://doi.org/10.4081/gh.2024.1271>.

- Kmoch, A., Harrison, C.T., Choi, J., Uemaa, E., 2025. Spatial autocorrelation in machine learning for modelling soil organic carbon. *Ecol. Inform.* 86, 103057. <https://doi.org/10.1016/j.ecoinf.2025.103057>.
- Kuang, Y., Chen, X., 2025. Spatial heterogeneity of forest carbon stocks in the Xiangjiang river basin urban agglomeration: analysis and assessment based on the multiscale geographically weighted regression (MGWR) model. *Front. Environ. Sci.* 13, 1573438. <https://doi.org/10.3389/fenvs.2025.1573438>.
- Kumar, S., Lal, R., Liu, D., 2012. A geographically weighted regression kriging approach for mapping soil organic carbon stock. *Geoderma* 189–190, 627–634. <https://doi.org/10.1016/j.geoderma.2012.05.022>.
- Kumar, A., Moharana, P.C., Jena, R.K., Malyan, S.K., Sharma, G.K., Fagodiya, R.K., Shabnam, A.A., Jigyasu, D.K., Kumari, K.M.V., Doss, S.G., 2023. Digital mapping of soil organic carbon using machine learning algorithms in the upper Brahmaputra Valley of northeastern India. *Land* 12, 1841. <https://doi.org/10.3390/land12101841>.
- Lal, R., 2004. Soil carbon sequestration impacts on global climate change and food security. *Science* 304, 1623–1627. <https://doi.org/10.1126/science.1097396>.
- Lamsaf, A., Carrilho, R., Neves, J.C., Proença, H., 2025. Causality, machine learning, and feature selection: a survey. *Sensors* 25, 2373. <https://doi.org/10.3390/s25082373>.
- Leong, Y.-Y., Yue, J.C., 2017. A modification to geographically weighted regression. *Int. J. Health Geogr.* 16, 11. <https://doi.org/10.1186/s12942-017-0085-9>.
- Li, S., Zhao, Z., Miaomiao, X., Wang, Y., 2010. Investigating spatial non-stationary and scale-dependent relationships between urban surface temperature and environmental factors using geographically weighted regression. *Environ. Model. Softw.* 25, 1789–1800. <https://doi.org/10.1016/j.envsoft.2010.06.011>.
- Ließ, M., Schmidt, J., Glaser, B., 2016. Improving the spatial prediction of soil organic carbon stocks in a complex Tropical Mountain landscape by methodological specifications in machine learning approaches. *PLoS One* 11, e0153673. <https://doi.org/10.1371/journal.pone.0153673>.
- Liu, Y., Wang, K., Dong, L., Li, J., Wang, X., Shangguan, Z., Qu, B., Deng, L., 2023. Dynamics of litter decomposition rate and soil organic carbon sequestration following vegetation succession on the loess plateau, China. *CATENA* 229, 107225. <https://doi.org/10.1016/j.catena.2023.107225>.
- Marsett, R.C., Qi, J., Heilman, P., Society for Range Management, 2006. Remote sensing for grassland management in the Arid Southwest. *REM* 59. [https://doi.org/10.2458/azu\\_jrm\\_v59i5\\_marsett](https://doi.org/10.2458/azu_jrm_v59i5_marsett).
- McBratney, A.B., Mendonça Santos, M.L., Minasny, B., 2003. On digital soil mapping. *Geoderma* 117, 3–52. [https://doi.org/10.1016/S0016-7061\(03\)00223-4](https://doi.org/10.1016/S0016-7061(03)00223-4).
- Minasny, B., McBratney, A.B., 2006. A conditioned Latin hypercube method for sampling in the presence of ancillary information. *Comput. Geosci.* 32, 1378–1388. <https://doi.org/10.1016/j.cageo.2005.12.009>.
- Minasny, B., McBratney, A.B., 2016. Digital soil mapping: a brief history and some lessons. *Geoderma* 264, 301–311. <https://doi.org/10.1016/j.geoderma.2015.07.017>.
- Minasny, B., McBratney, A.B., Malone, B.P., Wheeler, I., 2013. Digital mapping of soil carbon. In: *Advances in Agronomy*. Elsevier, pp. 1–47. <https://doi.org/10.1016/b978-0-12-405942-9.00001-3>.
- Mishra, U., Lal, R., Liu, D., Van Meirvenne, M., 2010. Predicting the spatial variation of the soil organic carbon pool at a regional scale. *Soil Sci. Soc. Am. J.* 74, 906–914. <https://doi.org/10.2136/sssaj2009.0158>.
- Mishra, U., Gautam, S., Riley, W.J., Hoffman, F.M., 2020. Ensemble machine learning approach improves predicted spatial variation of surface soil organic carbon stocks in data-limited northern circumpolar region. *Front. Big Data* 3. <https://doi.org/10.3389/fdata.2020.528441>.
- Moura-Bueno, J.M., Dalmolin, R.S.D., Horst-Heinen, T.Z., Grunwald, S., Ten Caten, A., 2021. Environmental covariates improve the spectral predictions of organic carbon in subtropical soils in southern Brazil. *Geoderma* 393, 114981. <https://doi.org/10.1016/j.geoderma.2021.114981>.
- Munoz, M.A., Faz, A., Mermut, A.R., 2015. Soil carbon reservoirs at high-altitude ecosystems in the Andean plateau. In: Öztürk, M., Hakeem, K.R., Faridah-Hanum, I., Efe, R. (Eds.), *Climate Change Impacts on High-Altitude Ecosystems*. Springer International Publishing, Cham, pp. 135–153. [https://doi.org/10.1007/978-3-319-12859-7\\_4](https://doi.org/10.1007/978-3-319-12859-7_4).
- Naimi, S., Ayoubi, S., Zeraatpisheh, M., Dematte, J.A.M., 2021. Ground observations and environmental covariates integration for mapping of soil salinity: a machine learning-based approach. *Remote Sens* 13, 4825. <https://doi.org/10.3390/rs13234825>.
- O'Brien, R.M., 2007. A caution regarding rules of thumb for variance inflation factors. *Qual. Quant.* 41, 673–690. <https://doi.org/10.1007/s11335-006-9018-6>.
- Otto, S., De Vos, B., Sindayihebura, A., Hermly, M., Van Orshoven, J., 2017. Assessing soil organic carbon stocks under current and potential forest cover using digital soil mapping and spatial generalisation. *Ecol. Indic.* 77, 139–150. <https://doi.org/10.1016/j.ecolind.2017.02.010>.
- Parastatidis, D., Mitraka, Z., Chrysoulakis, N., Abrams, M., 2017. Online global land surface temperature estimation from Landsat. *Remote Sens* 9, 1208. <https://doi.org/10.3390/rs9121208>.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., et al., 2011. Scikit-learn: machine learning in python. *J. Mach. Learn. Res.* 12, 2825–2830.
- Peng, Y., Chahal, I., Hooker, D.C., Van Eerd, L.L., 2024. Comparison of equivalent soil mass approaches to estimate soil organic carbon stocks under long-term tillage. *Soil Tillage Res.* 238, 106021. <https://doi.org/10.1016/j.still.2024.106021>.
- Pouladi, N., Gholizadeh, A., Khosravi, V., Borůvka, L., 2023. Digital mapping of soil organic carbon using remote sensing data: a systematic review. *CATENA* 232, 107409. <https://doi.org/10.1016/j.catena.2023.107409>.
- Priyatikanto, R., Lu, Y., Dash, J., Sheffield, J., 2023. Improving generalisability and transferability of machine-learning-based maize yield prediction model through domain adaptation. *Agric. For. Meteorol.* 341, 109652. <https://doi.org/10.1016/j.agrformet.2023.109652>.
- Pulgar Vidal, J., 2014. Las ocho regiones naturales del Perú. *Terra Bras.* <https://doi.org/10.4000/terrabrasilis.1027>.
- Pylaniadis, C., Kallenberg, M.G.J., Athanasiadis, I.N., 2024. Domain adaptation with transfer learning for pasture digital twins. *Environ. Data Sci.* 3, e8. <https://doi.org/10.1017/eds.2024.6>.
- R Core Team, 2023. R: A Language and Environment for Statistical Computing (Manual). R Foundation for Statistical Computing, Vienna, Austria. <https://www.r-project.org/>.
- Rasel, S.M.M., Groen, T.A., Hussin, Y.A., Diti, I.J., 2017. Proxies for soil organic carbon derived from remote sensing. *Int. J. Appl. Earth Obs. Geoinf.* 59, 157–166. <https://doi.org/10.1016/j.jag.2017.03.004>.
- Rikimaru, A., Roy, P.S., Miyatake, S., 2002. Tropical forest cover density mapping. *Trop. Ecol.* 43, 39–47.
- Román-Sánchez, A., Vanwalleghe, T., Peña, A., Laguna, A., Giraldez, J.V., 2018. Controls on soil carbon storage from topography and vegetation in a rocky, semi-arid landscapes. *Geoderma* 311, 159–166. <https://doi.org/10.1016/j.geoderma.2016.10.013>.
- Roudier, P., 2012. clls: Conditioned Latin Hypercube Sampling. <https://doi.org/10.32614/CRAN.package.clls>.
- Rouse, J.W., Haas, R.H., Schell, J.A., Deering, D.W., 1974. Monitoring vegetation systems in the Great Plains with ERTS. In: NASA. Goddard Space Flight Center 3d ERTS-1 Symp. vol. 1. Sect. A.
- Santos, F., Graw, V., Bonilla, S., 2019. A geographically weighted random forest approach for evaluate forest change drivers in the northern Ecuadorian Amazon. *PLoS One* 14, e0226224. <https://doi.org/10.1371/journal.pone.0226224>.
- Sartika, E., Suryani, A., 2020. Comparison of geographically weighted regression analysis and global regression on modeling the unemployment rate in west java. In: *Proceedings of the International Seminar of Science and Applied Technology (ISSAT 2020)*. Atlantis Press, pp. 472–478. <https://doi.org/10.2991/aer.k.201221.078>.
- Schönlau, M., Zou, R.Y., 2020. The random forest algorithm for statistical learning. *Stata J.* 20, 3–29. <https://doi.org/10.1177/1536867X20909688>.
- Schwanghart, W., Jarmer, T., 2011. Linking spatial patterns of soil organic carbon to topography — a case study from South-Eastern Spain. *Geomorphology* 126, 252–263. <https://doi.org/10.1016/j.geomorph.2010.11.008>.
- Song, M., Huang, Z., Chen, C., Li, X., Mao, F., Huang, L., Zhao, Y., Lv, L., Yu, J., Du, H., 2024. Multi-scale geographically weighted regression estimation of carbon storage on coniferous forests considering residual distribution using remote sensing data. *Ecol. Indic.* 166, 112495. <https://doi.org/10.1016/j.ecolind.2024.112495>.
- Sun, Y., Ao, Z., Jia, W., Chen, Y., Xu, K., 2021. A geographically weighted deep neural network model for research on the spatial distribution of the down dead wood volume in Liangshui National Nature Reserve (China). *iForest* 14, 353–361. <https://doi.org/10.3832/efor3705-014>.
- Szakács, G.G.J., Cerri, C.C., Herpin, U., Bernoux, M., 2011. Assessing soil carbon stocks under pastures through orbital remote sensing. *Sci. Agric. (Piracicaba, Braz.)* 68, 574–581. <https://doi.org/10.1590/S0103-90162011000500010>.
- Taghizadeh-Mehrjardi, R., Schmidt, K., Amirian-Chakan, A., Rentschler, T., Zeraatpisheh, M., Sarmadian, F., Valavi, R., Davatgar, N., Behrens, T., Scholten, T., 2020. Improving the spatial prediction of soil organic carbon content in two contrasting climatic regions by stacking machine learning models and resampling covariate space. *Remote Sens* 12, 1095. <https://doi.org/10.3390/rs12071095>.
- Tahmouresi, M.S., Niksokhan, M.H., Ehsani, A.H., 2024. Enhancing spatial resolution of satellite soil moisture data through stacking ensemble learning techniques. *Sci. Rep.* 14, 25454. <https://doi.org/10.1038/s41598-024-77050-0>.
- Thrift, N.J., Kitchin, R., 2009. *International Encyclopedia of Human Geography*. Elsevier, Amsterdam Boston.
- Triantakostas, D., Karakostas, A., 2025. Soil organic carbon monitoring and modelling via machine learning methods using soil and remote sensing data. *Agriculture* 15, 910. <https://doi.org/10.3390/agriculture15090910>.
- Tyralis, H., Papacharalampous, G., Langousis, A., 2019. A brief review of random forests for water scientists and practitioners and their recent history in water resources. *Water* 11, 910. <https://doi.org/10.3390/w11050910>.
- Vallejos-Torres, G., Gaona-Jimenez, N., Pichis-García, R., Ordoñez, L., García-Gonzales, P., Quinteros, A., Lozano, A., Saavedra-Ramírez, J., Tuesta-Hidalgo, J.C., Reategui, K., Macedo-Córdova, W., Baselly-Villanueva, J.R., Marín, C., 2024. Carbon reserves in coffee agroforestry in the Peruvian Amazon. *Front. Plant Sci.* 15. <https://doi.org/10.3389/fpls.2024.1410418>.
- Wang, K., Zhang, C., Li, W., 2012. Comparison of geographically weighted regression and regression kriging for estimating the spatial distribution of soil organic matter. *GISci. Remote Sens.* 49, 915–932. <https://doi.org/10.2747/1548-1603.49.6.915>.
- Wang, D., Li, X., Zou, D., Wu, T., Xu, H., Hu, G., Li, R., Ding, Y., Zhao, L., Li, W., Wu, X., 2020. Modeling soil organic carbon spatial distribution for a complex terrain based on geographically weighted regression in the eastern Qinghai-Tibetan plateau. *CATENA* 187, 104399. <https://doi.org/10.1016/j.catena.2019.104399>.
- Wang, L., Abramowitz, G., Wang, Y.-P., Pitman, A., Viscarra Rossel, R.A., 2024. An ensemble estimate of Australian soil organic carbon using machine learning and process-based modelling. *SOIL* 10, 619–636. <https://doi.org/10.5194/soil-10-619-2024>.
- Wei, X., Shao, M., Gale, W., Li, L., 2014. Global pattern of soil carbon losses due to the conversion of forests to agricultural land. *Sci. Rep.* 4, 4062. <https://doi.org/10.1038/srep04062>.
- Weku, W., Pramoddy, H., Widodo, A., Fitriani, R., 2022. Optimal bandwidth for geographically weighted regression to model the spatial dependency of land prices in

- Manado, North Sulawesi Province. Indones. Geogr. Environ. Sustain. 15, 84–90. <https://doi.org/10.24057/2071-9388-2019-154>.
- Wheeler, D.C., 2007. Diagnostic tools and a remedial method for collinearity in geographically weighted regression. Environ. Plan. A 39, 2464–2481. <https://doi.org/10.1068/a38325>.
- Wiesmeier, M., Urbanski, L., Hobbey, E., Lang, B., von Lützow, M., Marin-Spiotta, E., van Wesemael, B., Rabot, E., Ließ, M., Garcia-Franco, N., Wollschläger, U., Vogel, H.-J., Kögel-Knabner, I., 2019. Soil organic carbon storage as a key function of soils - a review of drivers and indicators at various scales. Geoderma 333, 149–162. <https://doi.org/10.1016/j.geoderma.2018.07.026>.
- Wu, S., Jia, W., Wang, F., Sun, Y., Zhao, H., Lu, S., 2023. Estimation of above-ground carbon storage and light saturation value in northeastern China's natural forests using different spatial regression models. Forests 14. <https://doi.org/10.3390/f14101970>.
- Yang, W., Deng, M., Tang, J., Luo, L., 2022. Geographically weighted regression with the integration of machine learning for spatial prediction. J. Geogr. Syst. 25, 213–236. <https://doi.org/10.1007/s10109-022-00387-5>.
- Yigini, Y., Panagos, P., 2016. Assessment of soil organic carbon stocks under future climate and land cover changes in Europe. Sci. Total Environ. 557–558, 838–850. <https://doi.org/10.1016/j.scitotenv.2016.03.085>.
- Zanaga, D., Van De Kerchove, R., Daems, D., De Keersmaecker, W., Brockmann, C., Kirches, G., Wevers, J., Cartus, O., Santoro, M., Fritz, S., Lesiv, M., Herold, M., Tsendbazar, N.-E., Xu, P., Ramoino, F., Arino, O., 2022. ESA WorldCover 10 m 2021 v200. <https://doi.org/10.5281/ZENODO.7254221>.
- Zeng, C., Yang, L., Zhu, A.-X., Rossiter, D.G., Liu, Jing, Liu, Junzhi, Qin, C., Wang, D., 2016. Mapping soil organic matter concentration at different scales using a mixed geographically weighted regression method. Geoderma 281, 69–82. <https://doi.org/10.1016/j.geoderma.2016.06.033>.
- Zeng, Y., Shi, T., Liu, Q., Yang, C., Zhang, Z., Wang, R., 2024. A geographically weighted neural network model for digital soil mapping of heavy metal copper in coastal cities. J. Hazard. Mater. 480, 136285. <https://doi.org/10.1016/j.jhazmat.2024.136285>.
- Zhang, Z., Jung, C., 2021. GBDT-MO: gradient-boosted decision trees for multiple outputs. IEEE Trans. Neural Netw. Learn. Syst. 32, 3156–3167. <https://doi.org/10.1109/TNNLS.2020.3009776>.
- Zhang, C., Tang, Y., Xu, X., Kiely, G., 2011. Towards spatial geochemical modelling: use of geographically weighted regression for mapping soil organic carbon contents in Ireland. Appl. Geochem. 26, 1239–1248. <https://doi.org/10.1016/j.apgeochem.2011.04.014>.
- Zhang, W., Wan, H., Zhou, M., Wu, W., Liu, H., 2022. Soil total and organic carbon mapping and uncertainty analysis using machine learning techniques. Ecol. Indic. 143, 109420. <https://doi.org/10.1016/j.ecolind.2022.109420>.
- Zhang, W., Ji, J., Li, B., Deng, X., Xu, M., 2025. Integrating genetic algorithm and geographically weighted approaches into machine learning improves soil pH prediction in China. Remote Sens 17, 1086. <https://doi.org/10.3390/rs17061086>.